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WORKING PAPER

ELECTRIFYING RIDE-HAILING IN THE UNITED STATES, EUROPE, AND CANADA: HOW TO ENABLE RIDE-HAILING DRIVERS TO SWITCH TO ELECTRIC VEHICLES

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EXECUTIVE SUMMARY

Highlights

- Governments' efforts to reduce transportation emissions through electrification can accelerate their impact by focusing on intensively used vehicles.
- Vehicles driven on ride-hailing platforms such as Uber and Lyft are often intensively used, but are overlooked by electric vehicle (EV) policies, and drivers face several barriers to acquiring EVs. An array of disparately available financial incentives, EV models, and charging options produce a complicated landscape where it is often unclear to potential EV buyers whether an EV costs more or less than an internal combustion engine (ICE) vehicle.
- As a result, in U.S., European, and Canadian cities, the share of EVs among vehicles used for ride-hailing is often lower than or similar to the share of EVs in the city's overall vehicle stock. However, new analysis of 10 cities shows that in places with enabling environments, including London and Vancouver, the electrification rate among vehicles used for ride-hailing is significantly higher than the city vehicle stock. This can accelerate progress toward cities' climate targets, support the development of charging infrastructure, and facilitate a more equitable transition to EVs.
- National and subnational governments, ride-hailing companies, utilities, and charging station operators, and other actors all have roles to play in facilitating the transition to electric ride-hailing and unlocking those benefits. Top-priority actions include improving the targeting of EV purchase subsidies, installing more

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350 kilowatt chargers and overnight charging for renters or residents of multiunit buildings, revising pricing structures for EV fast charging, and tailoring ride-hailing apps to support drivers with EVs.

Context: The Impacts and Status of Electric Ride-Hailing in the Transition to Sustainable, Equitable Mobility

Fighting climate change and air pollution requires shifting from ICE vehicles to EVs and increasing shared and active transportation and renewable energy generation. ICE vehicles emit greenhouse gases with each mile they drive; thus, larger emissions reductions would come from electrifying more intensively used (high-mileage) vehicles (Jenn 2020; McLane et al. 2021).

Vehicles used for ride-hailing are driven more than the average household vehicle (McLane et al. 2021; Pavlenko et al. 2019),[1] but because they are privately owned, they are often overlooked by actions to electrify other high-mileage, light-duty commercial vehicles such as taxis and delivery vehicles. In addition, EV adoption is lagging among people with lower incomes, who make up the majority of ride-hailing drivers (see Appendix B). This means that accelerating the electrification of vehicles used for ride-hailing is an opportunity for governments to both reduce greenhouse gas emissions and increase low-income drivers' access to EVs.

In many cities, this opportunity is currently being missed. In U.S., European, and Canadian cities, the share of EVs among vehicles used for ride-hailing is often lower than or similar to the share of EVs in the overall vehicle stock (BloombergNEF 2020).[2] However, new analysis of 10 cities shows that in places with enabling environments, the electrification rate among vehicles used for ride-hailing is significantly higher than the city vehicle stock, including London and Vancouver. This shows that a different approach is needed, but with concerted effort, national and subnational governments, ride-hailing companies, utilities, and charging station operators can support the transition to electric ride-hailing. Therefore, this paper provides guidance to aid a sustainable, equitable transition to the full electrification of vehicles used for ride-hailing in the United States, Europe, and Canada.

About This Paper

This paper explores the impacts and status of electric ride-hailing, describes four main barriers to electric ride-hailing, and presents actions that governments, ride-hailing companies, utilities, and other actors can take to overcome them, based on a literature review of academic and gray publications and consultations with experts and industry actors. It concludes with city scorecards that evaluate 10 cities on their progress towards dismantling these barriers, using an original methodology developed by the World Resources Institute and data shared by Uber. Given the urgent need to fully electrify and decarbonize transportation (IEA 2021; IPCC 2021), EVs are defined as being fully battery electric and fuel cell only, excluding others such as plug-in hybrids.

Key Findings and Possible Solutions

The higher up-front cost of EVs is a major barrier for many ride-hailing drivers, but shifting the targeting of current financial incentives and subsidies can cost-effectively mitigate this barrier until EVs reach price parity. Even though the relative total cost of ownership of EVs for ride-hailing is sometimes favorable and will likely be widely cost competitive by 2026, purchasing an EV often requires a larger up-front investment than a comparable ICE vehicle, which is out of reach for many ride-hailing drivers (Pavlenko et al. 2019; Rajagopal and Phadke 2019). This is exacerbated by public subsidies that often do not target the most intensely used vehicles or the people most in need of financial support (Bauer et al. 2021; Rajagopal and Phadke 2019) and by low-income households' lack of access to affordable auto loans (Grid Alternatives and EVNoire 2021; Milken Institute 2017). To overcome this barrier, governments at all levels can improve the targeting of their EV subsidies and incentives, such as by targeting them to intensively used vehicles, low-income households, or ride-hailing drivers. Governments can adjust taxes and fees to bolster the economic case for electric ride-hailing (Slowik et al. 2019c), and ride-hailing companies can offer financial support to drivers. There is also potential for innovation in the models under which vehicles are leased, rented, or shared to spread up-front costs over time or among drivers (Milken Institute 2017).

To transition to EVs, many ride-hailing drivers will need access to overnight charging at or near their homes, which is currently a challenge for many drivers who rent their homes or live in

multiunit buildings. It will also be important to expand access to public fast charging (Nicholas et al. 2020). Governments can increase overnight charging access by adding on-street charging in residential areas and by implementing "charge by right" mandates so renters can install home chargers (McLane et al. 2021), and ride-hailing companies can partner with utilities and parking management entities to offer drivers near-home overnight charging.

Policies to ensure access to overnight charging will also be essential to make EV charging affordable for ride-hailing drivers. Intensively used vehicles can have a stronger financial case for electrification because their operational savings offset their higher up-front costs more quickly—as long as owners have access to affordable charging (Bates and Leibling 2012; McLane et al. 2021; Rajagopal and Yang 2020). A major component of charging affordability is access to overnight charging; without it, drivers must rely on public chargers with higher electricity prices and time costs (Bauer et al. 2021). Utilities have a major role to play here, including implementing variable electricity prices (Chitkara et al. 2016; Fitzgerald and Nelder 2019). Governments can subsidize home charging equipment and installation, and ride-hailing companies can negotiate with utilities and charging station operators to secure favorable charging rates and access for ride-hailing drivers (McLane et al. 2021). They can also streamline and share information with drivers so they can make informed, price-responsive charging choices.

Governments and ride-hailing companies can work together to increase awareness of EVs and access to relevant information among ride-hailing drivers and low-income communities (Grid Alternatives and EVNoire 2021). Improving all drivers' baseline knowledge of EVs will make them more likely to consider one when the time comes to acquire a new vehicle, which is often an unexpected event that leaves little time for research and cost comparison (Grid Alternatives and EVNoire 2021; Milken Institute 2017). Ride-hailing companies can improve visibility and communication around EVs by creating EV educational materials that are appropriate for different population groups. Municipal governments can create a one-stop-shop webpage compiling all subsidies and programs available in that city (Milken Institute 2017).

Cities are taking a range of approaches to dismantle barriers to vehicle electrification, and some are advancing faster than others. Emerging trends from the city scorecards show that, in general, U.S. cities are lagging in their rates of ride-hailing electrification (1 percent or less), with Amsterdam, London, and Vancouver leading with rates over 4.5 percent (Uber 2021).

1. THE ROLE AND STATUS OF ELECTRIC RIDE-HAILING IN THE TRANSITION TO SUSTAINABLE, EQUITABLE MOBILITY

What Are the Impacts of Electric Ride-Hailing, and How Does It Fit into Sustainable, Equitable Mobility Systems?

Decarbonizing transportation can help governments achieve climate commitments and improve air quality. The transport sector accounts for almost a quarter of global carbon dioxide (CO_2) emissions (IEA 2020), and in many countries, the majority of these emissions come from passenger vehicles; in the United States, for example, passenger vehicles accounted for 58 percent of transportation emissions in 2019 (EPA 2019). In 2015, air pollution from transportation resulted in global health damages equivalent to 7.8 million years of life lost and approximately $1 trillion (Anenberg et al. 2019).

Electrifying intensively used vehicles will deliver climate benefits faster (Jenn 2020). Internal combustion engine (ICE) vehicles emit greenhouse gases with each mile they drive; as a result, larger emissions reductions would come more quickly from electrifying more intensively used (high-mileage) vehicles (McLane et al. 2021). The precise emissions reductions depend on the local electricity mix. In California in 2019, an electric vehicle (EV) used full-time for ride-hailing emitted 79 percent less CO_2 than a comparable ICE vehicle (Jenn 2019), and in Brazil, an electric taxi fleet would produce 10 times less CO_2 emissions than a conventional fleet due to Brazil's relatively clean energy mix (Teixeira and Sodré 2016).

Although vehicles used for ride-hailing are intensively used, they are often missed by current electrification policies. A full-time ride-hailing driver drives roughly three times more than the average American (McLane et al. 2021), and vehicles on Uber are generally driven more than those used for private use or car sharing (Pavlenko et al. 2019).[3] Vehicles used for ride-hailing are driven up to four times more than the average EV (Rajagopal and Yang 2020). However, since vehicles used for ride-hailing are privately owned, they are missed by actions aiming to electrify other high-mileage, light-duty commercial vehicles such as taxis and delivery vans. This is reflected in real-world adoption: in U.S., European, and Canadian cities, the share of EVs among vehicles used for ride-hailing is often lower than or similar to the share of EVs in the overall vehicle stock (BloombergNEF 2020).[4] In addition, many vehicles are used for ride-hailing only temporarily and part-time (Klein 2020; Mischel 2018), and EV adoption is lagging among people with the demographic profile of typical ride-hailing drivers, who often come from the lower half of the income distribution, with an average annual household income below US$50,000 (see Appendix B, "Estimating Ride-Hailing Drivers' Income"). These factors mean that a different approach is needed to accelerate the transition to electric ride-hailing. Therefore, this paper provides guidance to aid a sustainable, equitable transition to electric ride-hailing.

Electric ride-hailing must be considered in the wider context of a sustainable, equitable transportation system. From a climate perspective, the most important goal is reducing average CO_2 emissions per passenger mile (including miles traveled on nonmotorized modes and public transit) while reducing the total number of passenger miles. According to the Avoid-Shift-Improve framework, this is best achieved by a tiered approach:

- **Avoid:** Reduce the need for travel through compact, accessible urban planning.
- **Shift:** Shift travel to walking, cycling, and shared transport as much as possible while reducing share of travel in private vehicles
- **Improve:** Ensure that remaining passenger miles are in electric and efficient vehicles (Dalkmann and Brannigan 2007)

Within that framework, electric ride-hailing falls within the third tier. It can also fall within the second tier to the extent that ride-hailing facilitates increased vehicle occupancy, more multimodal trips, or reduces reliance on private vehicle ownership. Therefore, electric ride-hailing's contribution to decarbonizing transportation depends on the extent to which it replaces (rather than adds to) private vehicle miles and ICE ride-hailing miles, which is a subject of debate (Xing et al. 2019). Evidence is complicated; transit riders are more likely than others to use ride-hailing, and increased ride-hailing use is associated with higher transit ridership, but some ride-hailing trips replace transit trips or generate new trips (TransitCenter 2019). Ride-hailing also adds to vehicle miles when deadheading between passengers, and the frequency of shared rides varies (but has dropped since COVID-19). Meanwhile, ride-hailing makes up only a small fraction of vehicle miles in most cities, so the overall scale of climate benefits is limited.

Prioritizing the electrification of ride-hailing (and other high-mileage, for-hire, and/or commercial vehicles) can accelerate and smooth a city's wider transition to electric transportation. Vehicles used for ride-hailing are more likely to charge at off-peak times (Jenn 2019), filling valleys or dampening peaks in electricity demand and enabling better grid load management. Renewable electricity is most abundant and affordable at off-peak daytime hours, so ride-hailing EVs can reduce a city's electricity storage needs and electricity-related emissions (Jenn 2020). Since they often use more public fast charging than personal vehicles (Jenn 2019), they can serve as "anchor tenants" to improve the early business case for public fast chargers, helping expand charging infrastructure and increase charging options for private EVs, especially in underserved communities (McLane et al. 2021). Electric ride-hailing can also expose more people to EVs, improving the public perception and eventual adoption of EVs, though further research is needed (Jenn et al. 2018; McLane et al. 2021). From an individual perspective, intensively used vehicles offer a stronger financial case for electrification because fuel savings more quickly offset EVs' higher up-front costs (Bates and Leibling 2012; McLane et al. 2021; Rajagopal and Yang 2020).

For transportation policies to be truly sustainable, they must prioritize equity. This means ensuring that resulting economic, social, and environmental benefits (including access to opportunity; see Venter et al. 2019) are distributed equitably among households and neighborhoods of different income levels and backgrounds, and that the costs of the transition are distributed fairly and account for historical and ongoing injustices in transportation access and public investments. It also means equitably allocating public and private spending among different populations. If subsidies and policies are not targeted to benefit low-income households (such as ride-hailing drivers), there is potential for regressive impacts. For example, in the United States, households earning over $150,000 account for 35 percent of purchases of hybrids and battery electric vehicles (BEVs), and those earning less than $50,000 account for just 14 percent, meaning that most public spending on EV subsidies benefits higher-income households (Muehlegger and Rapson 2018). Low-income and minority communities often have less access to EV charging (Hsu and Fingerman 2021). In addition, narrowly focusing only on electrification can risk entrenching passenger vehicles in mobility systems, even though shifting to nonmotorized and shared transportation would better promote equity (Bauer et al. 2021).

If well designed, policies to support electric ride-hailing could have equity benefits. Promoting electric ride-hailing could benefit low-income households because those same policies could make EVs more affordable. In the United States, transportation is often the second-largest household expenditure (after housing), largely due to the high fixed costs of vehicle ownership, which disproportionately burden low-income households. Since low-income households often buy cheaper, older cars that break down more and have worse fuel economy, the fuel and maintenance savings of EVs could be especially large for low-income households once the up-front vehicles costs are affordable. Replacing ICE vehicles with EVs in low-income communities would help alleviate their disproportionately high air pollution (Hajat et al. 2015). Since the transition to sustainable, equitable mobility requires a shift away from private cars and towards shared, active modes, it may seem that promoting any car ownership—including in low-income communities or for EVs—may work at cross-purposes with that goal. However, during the decades needed to transition to a more sustainable, equitable mobility system, many low-income households will still be compelled to buy vehicles and will continue to be burdened by transportation costs. Strategically targeted policies to promote EV adoption by low-income households can help reduce the financial burden and pollution that these communities experience in the near term, if they work alongside policies and infrastructure investments for shared, active mobility (King et al. 2019; Klein 2020).

Therefore, this paper analyzes actions related to electric ride-hailing based on the extent to which they advance the following goals:

- Reducing the average CO_2 equivalent (CO_2e) emissions per passenger mile (on any mode, including walking, cycling, micromobility, and mass transit) while reducing the total number of passenger miles traveled (CO_2e/pmt)
- Maximizing reductions in CO_2 emissions per unit of public (and private) spending (CO_2e/$)
- Allocating funds in a fair way that creates equitable outcomes

About This Paper

This paper provides guidance to aid the electrification of vehicles used for ride-hailing to deliver the benefits described in the preceding section. Section 1 describes the climate and equity benefits of electric ride-hailing and the importance of integrating into a sustainable, equitable mobility system. It describes the status of the transition to electric ride-hailing and the policy-relevant characteristics of ride-hailing drivers and today's EV owners. Section 2 describes the four main barriers to electric ride-hailing and suggests actions that a range of actors can take to overcome them. Section 3 layers a temporal element onto those barriers and actions, outlining the phases on the path towards electric ride-hailing. These sections are primarily based on a literature review on topics including EV policy, ride-hailing, sustainable mobility systems, and transportation equity. Places vary in their approach to and success in electrifying ride-hailing, so the city scorecards offer a snapshot of the readiness of and adoption by 10 cities, drawing on a wide range of city-specific data sources as well as proprietary data from Uber. Section 4 describes the trends that emerge from this analysis.

In this paper, EVs are defined as being fully battery electric and fuel cell only, excluding others such as plug-in hybrids. Achieving the goals of the Paris Agreement will require reaching net-zero emissions by 2050 (IPCC 2021). According to International Energy Agency models, this means that global CO_2 emissions from light-duty vehicles should have already peaked and should cease entirely by the late 2040s (IEA 2021). From a just transition perspective, developed countries should reduce their emissions fastest (Muttitt and Kartha 2020). This highlights the urgency of transitioning to fully electric vehicles (powered by a clean energy supply) in the places that are the focus of this paper: the United States, Europe, and Canada. Although hybrids have a role in the transition, analysis of their impacts and the actions that specifically support their adoption are out of scope for this paper.

This paper applies to ride-hailing in the United States, Europe, and Canada, facilitated by any company. However, the authors had access to proprietary data from Uber, so many examples and statistics refer to Uber specifically. The trends documented for Uber likely apply to other ride-hailing companies because many trends are primarily due to external economic and policy conditions. However, data from other companies would be needed to confirm this, and EV adoption may vary based on each company's EV-related initiatives.

What Is the Status of the Transition to Electric Ride-Hailing?

In the United States, EVs are more likely to be used for personal driving than ride-hailing; in Europe, trends are reversed. In the United States, the share of EVs among vehicles used for ride-hailing was lower than the share of EVs in the overall passenger vehicle stock in 2019 (see Table 1). In Europe, the U.S. pattern was reversed: three times more vehicles used for ride-hailing than passenger vehicles were electric.

Table 1 | **Share of EVs among Vehicles Used for Ride-Hailing and Private Passenger Vehicles, 2019**

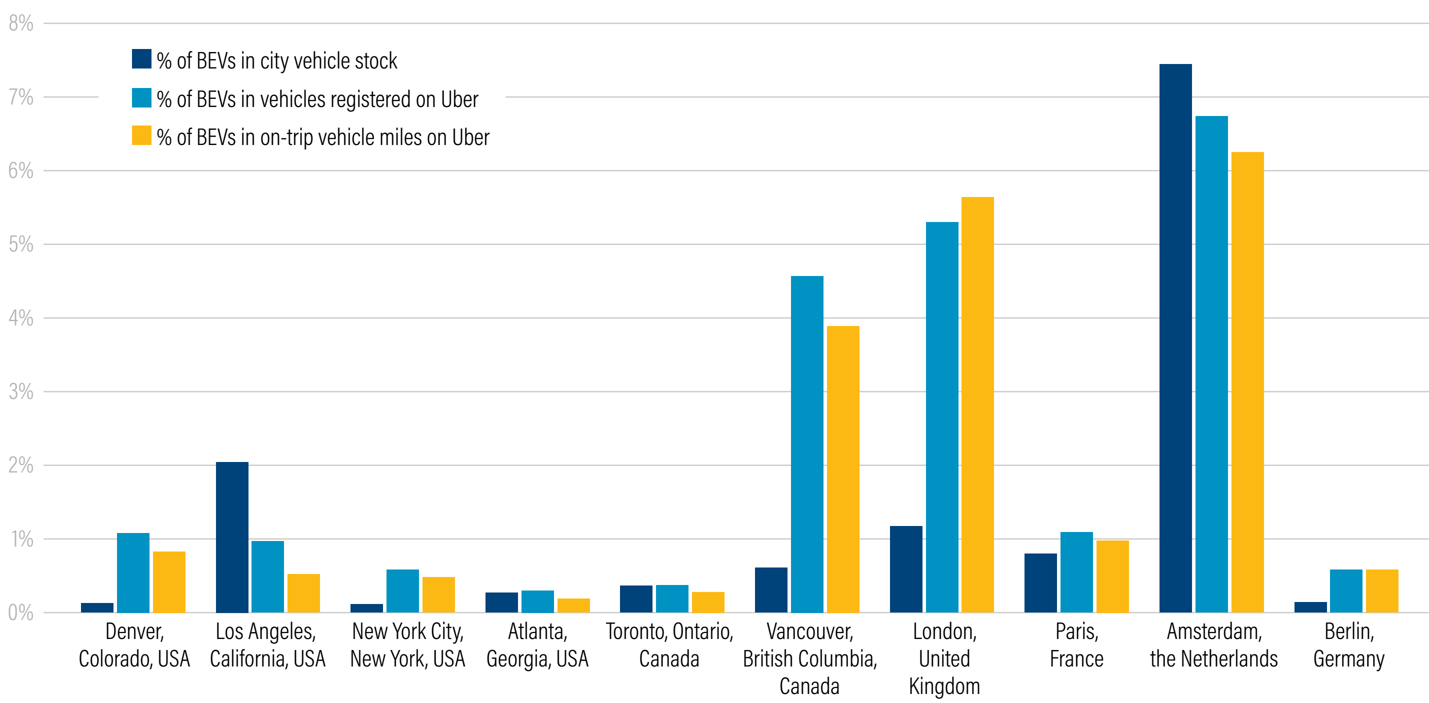
SHARE OF EVS	UNITED STATES (%)	EUROPE (%)
Vehicles used for ride-hailing	0.3	2.1
Private passenger vehicles	0.6	0.7

Note: EV = electric vehicles.

Source: BloombergNEF 2020.

However, momentum is building. In the private sector, Uber has set a target of 100 percent electrification in U.S., European, and Canadian cities by 2030 and becoming a fully zero-emissions platform by 2040 (Uber 2020a), and Lyft has committed to 100 percent electrification by 2030 (Lyft 2020), though there is still a long way to go (see Figure 1). Vehicle manufacturers announcing phaseout targets for ICE vehicles, such as General Motors' goal to sell only zero-emissions vehicles by 2035, and EV costs are falling (Boudette and Davenport 2021). Many countries, states, and cities have targets and policies to accelerate EV adoption, including some city policies to promote electric ride-hailing (Sheldon and Dua 2019).

Figure 1 | **Share of EVs among City Vehicle Stock, Vehicles Registered on Uber, and On-Trip Vehicle Miles on Uber, January-June 2021**



Notes: BEV = battery electric vehicle. On-trip miles are miles for which the vehicle contained one or more passengers.

Sources: DOE 2021b; Dunsky Energy Consulting 2019; Government of BC 2021; Hall et al. 2020; Uber 2021.

2. BARRIERS TO ELECTRIC RIDE-HAILING AND ACTIONS THAT CAN OVERCOME THOSE BARRIERS

This section is structured around four major barriers to electric ride-hailing:

- **Barrier 1. Vehicle purchase and ownership costs:** EVs are often more expensive than ICE vehicles.
- **Barrier 2. Charging access:** Some ride-hailing drivers lack convenient EV charging options.
- **Barrier 3. Charging costs:** Some ride-hailing drivers lack affordable EV charging options.
- **Barrier 4. Awareness and access:** Some ride-hailing drivers lack awareness or access to information regarding EVs.

The section for each barrier describes its subcomponents, magnitude, and impacts. It then offers potential actions that governments, ride-hailing companies, and other actors can take to overcome that barrier.

Methodology for Selecting and Grouping Barriers and Actions

Research for this paper was primarily based on a literature review that included publications from academic journals, think tanks and nonprofits, and select industry publications, media articles, or gray literature. Search terms included *electric* (and variants such as *electrification* or *electrify*) and *decarbonize* (with variants and British spellings) in combination with *ride-hailing*, *ridesharing*, *transportation network company* (TNC), and *Uber*, *Lyft*, and other TNCs.

Addressing all possible barriers to electric ride-hailing was beyond the scope of this publication. To select which barriers to include, the authors drew on the literature review to assess the extent to which a barrier met the following criteria:

- Is the barrier identified in the literature as a barrier that currently limits the adoption of EVs for ride-hailing?
- Can the barrier be significantly reduced by changes in policies and investments rather than technological or macroeconomic changes?
- Is it a barrier that is primarily faced by ride-hailing drivers (rather than governments, manufacturers, etc.)?

Based on these criteria, the authors excluded the following barriers, though they remain important for future research:

- Insufficient range was not addressed as a standalone barrier because it consists of elements that were covered in the four selected barriers. It is a financial barrier to the extent that batteries with larger capacities are more expensive (Barrier 1), and it can also be a function of insufficient charging infrastructure, especially public fast charging (Barrier 2). Limited model availability in certain locations is addressed in Barrier 4. Barriers related to battery technology would be excluded under the second criterion. Moreover, some literature indicates that many currently available EVs have sufficient range for ride-hailing (Sanguinetti and Kurani 2020).
- Limited availability of models that meet taxi or TNC standards was excluded because the literature did not indicate that this barrier currently significantly limits the adoption of EVs for ride-hailing.
- High up-front investment for public chargers was excluded because this barrier is not primarily faced by ride-hailing drivers.
- Lack of coordination on smart charging and vehicle-to-grid networks was excluded because the literature did not indicate that this barrier currently significantly limits the adoption of EVs for ride-hailing given the small potential financial benefits (roughly $100 per vehicle per year; Sovacool et al. 2017).
- Price discrimination by dealerships against low-income buyers was excluded because the literature did not indicate this occurs regularly (Muehlegger and Rapson 2018).

This paper grouped together barriers that would be best addressed by similar actions. For example, although charging costs and charging access are both related to charging, a primary way of addressing costs is through electricity pricing structures, and a primary way of improving access is by expanding charging infrastructure.

In determining how to address these barriers, the authors compiled potential actions from the literature review (as described above) and from consultations within the World Resources Institute (WRI) and close partners. The long list of potential actions was narrowed down to the actions included in this publication based on criteria that included alignment with the Avoid-Shift-Improve framework and equity goals; applicability in U.S., European, and Canadian contexts; the scale of the potential impact; and political feasibility. Sources for specific recommendations are cited where applicable.

Barrier 1. Vehicle Purchase and Ownership Costs: EVs Are Often More Expensive than ICE Vehicles

EVs are often unaffordable to ride-hailing drivers

As of 2021, both the up-front cost and total cost of ownership (TCO) are generally higher for EVs than for ICE vehicles (BloombergNEF 2020; Pavlenko et al. 2019; Rajagopal and Phadke 2019), typically by several thousands of dollars in 2018 (Pavlenko et al. 2019). Even when the TCO is lower for EVs, the payback period may be five years or more (Rajagopal and Phadke 2019). Meanwhile, ride-hailing drivers often come from the lower half of the income distribution, with an average annual household income below $50,000 (see Appendix B, "Estimating Ride-Hailing Drivers' Income"). Data on drivers' earnings are limited and are complicated by the fact that many drivers work part-time, temporarily (an average of three months), and have other income sources (Grid Alternatives and EVNoire 2021; Klein 2020; Mischel 2018). Still, ride-hailing drivers' lower household income likely explains why they often cite the up-front cost as their main barrier to purchasing an EV (Grid Alternatives and EVNoire 2021; Sanguinetti and Kurani 2020).

Ride-hailing drivers may also lack access to affordable financing options and credit. People with low incomes, low or no credit scores, or existing debt burdens may not qualify for car loans or lease-to-purchase programs, or they may only qualify for loans at high interest rates (Milken Institute 2017; Rajagopal and Phadke 2019). This means that an EV's TCO is often higher for lower-income households than for high-income households: African American and Latinx consumers sometimes pay $5,000 more than other consumers for the same vehicle (Grid Alternatives and EVNoire 2021). Lower-income households also tend to pay for cars in cash, meaning that they need to have saved the full vehicle cost rather than amortizing costs like many higher-income buyers (Milken Institute 2017).

The cost and availability of used EVs are relevant for electrifying ride-hailing. Most cars purchased by lower-income households are used (Bauer 2021), and anecdotal evidence suggests that many vehicles driven for Uber are purchased used.[5] The average age of a vehicle when it is first driven for Uber is 2.4 years in Canada, 3.4 in the United States, and 3.9 in the United Kingdom.[6] Table 2 shows that lower-income households do not buy used vehicles at dramatically higher rates than higher-income households, but their vehicles tend to be older than the ones purchased by higher-income households and they tend to be more affordable models (Bauer et al. 2021), pointing to the need for a deeper used EV market. Still, as of 2020, EVs accounted for just 9 percent of listings on used vehicle sales websites (BloombergNEF 2020). Many of these are likely older models with limited ranges and top charging speeds (Voelcker 2021), making them a poor fit for ride-hailing. In most places, used EVs do not qualify for EV subsidies.

However, this is a short- to medium-term problem: the price gap between used EVs and used ICE vehicles is already smaller than between comparable new vehicles (Milken Institute 2017). New EVs and ICE vehicles are projected to reach price parity in 2026, with used EVs and ICE vehicles reaching price parity just two years later (Bauer et al. 2021), and the payback period shortening to two years by 2025 (Pavlenko et al. 2019). The high level of adoption of hybrid vehicles for ride-hailing (which was 25 percent in Los Angeles and Vancouver, 30 percent in Paris, 58 percent in London, and 66 percent in Berlin on Uber as of June 2021 [Uber 2021]) indicates that drivers will adopt sustainable vehicle technologies when they are cost competitive.

Current policies are poorly targeted

The lagging pace of electrification of vehicles used for ride-hailing is partly because many current EV subsidies are not strategically targeted, for example, at high-mileage vehicles or low-income households. This means that the subsidies do not maximize vehicle miles electrified or

Table 2 | **Summary of Household and Vehicle Statistics by Income Group in the United States**

	HOUSEHOLD INCOME				
	< $25,000	$25,000-$50,000	$50,000-$75,000	$75,000-$150,000	>$150,000
Vehicles owned	1.1	1.7	2.1	2.4	2.5
Average vehicle age	12.6	11.3	10.6	9.4	8.2
Average vehicle mileage (miles/year per vehicle)	10,300	10,700	11,100	11,600	11,700
Average price for vehicles when new	$32,800	$33,000	$33,700	$34,600	$38,000
Average vehicle age at purchase	7.2	6.0	5.5	4.7	3.9
Proportion of vehicles purchased new	33%	33%	32%	38%	50%
Duration of vehicle ownership before sale	5.3	5.7	5.9	6.1	6.4
Average vehicle fuel economy (miles/gallon)	21.7	22.0	22.1	22.6	23.0
Proportion of homeowners	35%	56%	69%	79%	87%
Population density (people per square mile)	5,490	4,711	4,721	4,842	5,973
Proportion of single-family homes	55%	62%	70%	77%	77%
Average charging rate ($/kWh)	0.173	0.170	0.165	0.163	0.168

Note: All dollar amounts are in U.S. dollars.

Source: Bauer et al. 2021.

social benefits per dollar. Today, many EV subsidies are per vehicle—regardless of the intensity of vehicle use or income level of the buyer—and only for new vehicles. Examples include the U.S. federal tax credit of up to $7,500 on the purchase of a new EV or plug-in hybrid (Fueleconomy.gov 2021), Canada's CAN$5,000 incentive for new battery electric, hydrogen, or hybrid vehicles (Transport Canada 2020), and the United Kingdom's £3,500 plug-in car grant (Gov.UK n.d.).

Targeting EV subsidies to lower-income households would have resulted in larger emissions reductions and more equitable outcomes. In the United States, 70 percent of the EV tax credits went to households that would have bought an EV anyway. These EVs replaced vehicles with above-average fuel efficiency (Xing et al. 2019), and they were often a second or third vehicle that would be driven less and/or replace a more fuel-efficient vehicle (Bauer et al. 2021; Rajagopal and Phadke 2019). Meanwhile, lower-income households often own older, fewer, and less fuel-efficient vehicles, so they use each one more intensively, even if they are not a ride-hailing driver. Lower-income households also have more financial need for the subsidy, and targeting EV subsidies to them could help alleviate air and noise pollution in disadvantaged communities (Rajagopal and Phadke 2019). In California, households in disadvantaged communities captured only 6 percent of the rebates for BEVs, in part because they may not have owed enough taxes to take advantage of the full tax credit (Rajagopal and Phadke 2019). There are growing calls to target policies to lower-income households or high-mileage vehicles (Rajagopal and Phadke 2019; Sheldon and Dua 2019; Xing et al. 2019). An example is Greece's EV subsidy for taxi owners, which is 33 percent higher than the subsidy for individuals (WFW 2020).

Actions to Address Barrier 1

Make EVs more affordable, especially for low-income and/or high-mileage drivers

ACTIONS BY RIDE-HAILING COMPANIES

- Offer direct financial support to drivers to purchase EVs, which could be on a per-trip or per-mile basis (as Uber does in London) or contingent on continued work for a certain duration or mileage. This could potentially be funded by tradable EV carbon offset credits.
- Advocate to national governments for improved targeting of EV subsidies or offer to match national or state government subsidies if a certain percentage of their drivers switch to EVs.
- Partner with financial institutions to offer low-income drivers affordable loans to purchase new or used EVs. Drivers' history with ride-hailing can support their loan application, possibly counteracting a low credit score.

ACTIONS BY GOVERNMENT

All national governments could

- coordinate incentives and subsidies from all levels of government to improve accessibility; and
- improve the targeting of EV subsidies (alternative approaches to targeting EV subsidies to electrify ride-hailing are analyzed in Table 3).

Table 3 | **Alternative Approaches to Targeting EV Subsidies to Electrify Ride-Hailing**

POLICY TARGETING	ANALYSIS	
	PROS	CONS
Lower-income households	▪ Public spending benefits lower-income households ▪ Local environmental and health benefits would accrue in low-income areas ▪ Easier to implement because income is relatively static and verifiable ▪ Likely to replace fuel-inefficient vehicles ▪ Could stimulate development of more affordable electric vehicle (EV) models and charging infrastructure in low-income areas	▪ Still a per-vehicle subsidy, not targeting intensively used vehicles ▪ EV uptake in low-income areas may remain limited by lack of chargers and lack of awareness of EVs
High-mileage or for-hire vehicles	▪ Maximizes emissions reductions ▪ Avoids appearance of public funds being used to benefit a private company ▪ Eligibility could be verified by mileage recorded on ride-hailing apps	▪ Could incentivize more driving
Used vehicles	▪ Would enable subsidies to apply to a larger share of all EV purchases ▪ Would lower the current EV price floor in most places	▪ Not likely to be the most effective way to accelerate EV uptake among low-income households or ride-hailing drivers because all households purchase used vehicles at relatively similar rates, and used EVs depreciate quickly[a]
Ride-hailing drivers	This could include a refundable tax credit to EV drivers based on reported in-app mileage. An up-front subsidy would more fully offset up-front cost barriers, but an associated requirement for continued ride-hailing driving could create challenges for drivers, such as if they need to stop driving due to a move, personal or health challenges, or changes in ride-hailing company policies. ▪ Most directly targets ride-hailing ▪ Indirectly targets both high-mileage and low-income drivers	▪ Potentially controversial for public funds to support electrification of vehicles used for ride-hailing but separate from other high-mileage, for-hire, or commercial vehicles

Note: a. Bauer et al. 2021.

Source: WRI authors.

The U.S. government could additionally

- revise the subsidy cap for all automakers, possibly by removing it entirely so it will continue for the most popular EV models, or by removing it for any EVs below an affordable price point;
- increase the federal tax deduction for commercial or small business vehicles that are electric, with a sunset clause; and
- replace the tax credit model of the federal subsidy with a point-of-sale benefit to enable all income levels to fully benefit, such as in Germany (Lambert 2020).

State/regional/provincial or municipal governments could

- offer rebates, tax credits, or other subsidies in addition to national incentives, which could apply to used EVs and use similarly improved targeting.

Governments at varying levels (depending on the country) could

- offer low-income drivers affordable loans or support with up-front EV payments, similar to government programs for mortgages and down payments on homes (Milken Institute 2017); and
- exempt EVs from registration fees (potentially replacing it with alternative funding sources such as road pricing), which phases out after a defined period or threshold of electrification.

ACTIONS BY OTHER ACTORS

- Multibank community development corporations could pool funding to spread risk and lend in underserved areas (Milken Institute 2017).
- Financial institutions making auto loans could offer interest buy-downs to make loans on EVs more affordable (Milken Institute 2017).
- A range of actors could develop innovative business models and improve technologies for battery reuse, lowering the cost of otherwise new vehicles while reducing EV lifecycle emissions. Vehicle manufacturers could implement battery buy-back programs or develop circular supply chains (partnering with specialized companies,[7] government initiatives,[8] or public-private collaborations[9]), or governments or utilities could "own" batteries to pool risk and ensure second-life uses.

Make ICE vehicles more expensive to drive than EVs

ACTIONS BY RIDE-HAILING COMPANIES

- Offer financial incentives to drivers with EVs. However, in the short term, these will accrue mainly to drivers with above-average incomes, so they could be made available only to drivers of certain income levels or who live in designated areas. To avoid privacy concerns, drivers can opt-in to provide the personal information needed to determine eligibility.
- Create mechanisms for drivers to capitalize on riders' preference for sustainable vehicles, such as by expanding programs like Uber Green or letting customers indicate a preference for EVs (with the same caveat as above).

ACTIONS BY GOVERNMENTS

Vehicles used for ride-hailing are often subject to taxes and fees levied by governments, airports, and other entities as well as general road use-related fees such as congestion charging. These taxes and fees are often applied equally to all vehicles used for ride-hailing. Waiving or reducing them for EVs can incentivize electrification by lowering operating costs, boosting net revenue, and accelerating the break-even point with ICE vehicles. If electric vehicles used for ride-hailing were exempt from fees, EVs could become the most cost-effective option for ride-hailing in areas with high fees (such as New York City), even under the most expensive charging scenarios (Slowik et al. 2019c). However, it will be imperative to avoid undermining the reasons why these taxes and fees were implemented—namely, to incentivize the shift to shared, active modes, which are the foundations of sustainable, equitable mobility systems. Moreover, this approach assumes that taxes and fees are borne by drivers, but if a reduction in taxes or fees is passed on to ride-hailing customers in lower fares, it could incentivize more vehicle travel (Slowik et al. 2019c, 201).

Meanwhile, these revenues are often earmarked to fund projects such as sustainable transport, roadway improvement, or mobility for passengers with disabilities (Slowik et al. 2019c); however, in reality, the revenues are often not spent this way (Zhao et al. 2020). Nonetheless, it will be important to ensure that decreased revenue will not squeeze public budgets. The level of forecasted forgone revenue for U.S. cities varies from $1,000 to tens of thousands of dollars per vehicle over five years (Slowik et al. 2019c). Fee and tax structures could be revenue neutral, include a sunset date, and be lowered to the threshold where EVs become more economical than ICE vehicles (periodically adjusted as this threshold shifts). Since it can be politically challenging to remove subsidies, policies could include a sunset clause tied to an electrification or economic threshold. In London, for example, EVs lose their congestion charge exemption in December 2025, at which time Uber is targeting 100 percent electrification in London (Slowik et al. 2019c). Governments could set conditions for ride-hailing companies where they are granted exemptions to taxes and fees, such as sharing data or serving as paratransit in under-served areas.

- State/regional/provincial or municipal governments: Waive or reduce ride-hailing-related taxes and fees for EVs, or for fleets or companies with a certain share of EVs, with data reporting requirements for ride-hailing companies benefiting from these policies. Examples include San Francisco's reduction of the "TNC tax" by 50 percent for shared and EV trips (SFCTA n.d.) and London's exemption of EVs from congestion charge zone fees until 2025 (TfL n.d.a). For a detailed analysis, see Slowik et al. (2019c).
- National or state/regional/provincial governments: Grant cities the authority to implement policies and pricing schemes to promote EV uptake.

Innovate models for lease, rental, sharing, or ownership of vehicles used for ride-hailing to spread costs over time or among drivers

Many vehicles are only used for ride-hailing for a fraction of their life spans, complicating efforts to reduce EVs' up-front costs for ride-hailing drivers. Expanding options for drivers to lease, rent, or share EVs would let them use those EVs only during their duration as a ride-hailing driver and would make ride-hailing work accessible to people who do not own vehicles. If vehicles used for ride-hailing were owned and managed as a fleet, companies (either ride-hailing companies or other fleet owners) could save costs by optimizing charging, maintenance, and utilization and by negotiating lower prices on bulk vehicle purchases. These savings could be passed to drivers, increasing affordability, and these vehicles would eventually expand the used EV market. Centralizing fleet ownership could also enable partnerships with auto manufacturers to create custom ride-hailing models, such as the 2018 partnership in China between Didi and BYD (Hawkins 2020; Slowik et al. 2019b). In addition, as autonomous vehicles become more commercially viable, ride-hailing companies may shift to owning autonomous electric fleets (Iyer and Alton 2019; Shared-Use Mobility Center n.d.; Spencer 2021).

A growing number of examples indicate the viability of this model. General Motor's Maven program in 2017 was an early example of renting a fleet of EVs to ride-hailing drivers (Slowik et al. 2019b). A growing number of companies rent vehicles to ride-hailing drivers, mainly in the United States, including Lyft's Flexdrive (some EVs), Drive Sally (all EVs or hybrids), Buggy (no EVs as of August 2021), and Uber's partnership with Flux Auto, Hive, Avis, WeFlex, and Otto.[10] In India, Uber partnered with Lithium, an all-electric vehicle fleet operator, to offer 1,000 EVs on Uber Rentals and Premium (Uber 2020b).

ACTIONS BY GOVERNMENTS

- Create the policy conditions to operationalize these business models and regulate them appropriately.

ACTIONS BY RIDE-HAILING COMPANIES

- Negotiate favorable rates for drivers to rent EVs from vehicle rental companies.
- Rent or lease EVs to ride-hailing drivers or partner with companies that do so.

ACTIONS BY OTHER ACTORS

- Private companies could rent vehicles to ride-hailing drivers, moving towards fully electric fleets and expanding to other countries.
- Nonprofits, community development financial institutions, or other local organizations could experiment with community-owned fleets (Grid Alternatives and EVNoire 2021).

Barrier 2. Charging Access: Some Ride-Hailing Drivers Lack Convenient EV Charging Options

Lack of access to at-home overnight charging

To switch to an EV, drivers need to be able to charge their vehicles. Uber's pilot in London found that insufficient public charging was the main obstacle to EV adoption, and data from California indicate that ride-hailing EVs used public chargers more than personal EVs (Nicholas et al. 2020). There are two main methods for EV charging: at-home overnight charging and public fast charging. In many ways, at-home overnight charging is preferable because electricity costs are lower when vehicles can charge off-peak and more slowly, and drivers avoid the logistical challenges and time cost of charging during their ride-hailing sessions. Overnight charging also increases the time that vehicles are available for smart charging and vehicle-to-grid uses, which can help manage grid loads and further lower costs (Davar 2020). A lack of at-home charging was a main reason among EV owners (in California) who did not buy another EV for their subsequent vehicle (Hardman and Tal 2021). Therefore, a primary way to improve access is to expand access to at- or near-home overnight charging.

However, at-home overnight charging is unavailable to many drivers, especially those who live in a multiunit building, lack private parking, rent their home, or move frequently. Laws regarding the installation of EV chargers in rental properties are often unclear and differ on a city-by-city basis. For example, Oregon's legislation states that property owners can install home EV chargers for personal use, but it is unclear whether tenants of rental properties with private parking can also do so.[11] Buildings' physical limitations can exacerbate this barrier, especially inadequate electrical capacity in older buildings (Balmin et al. 2012). Even when laws are clear, the process is often complex, involving permits from entities such as a municipal utility district or a homeowners association as well as contacting the local utility company to determine the right equipment and installation process (Gurskiy 2019).

Lack of access to public charging

For some drivers, limited vehicle range or high daily mileage means they will need to use a public charger in addition to overnight charging. Among U.S. ride-hailing drivers, 24 percent of drivers say they are not willing to charge during their driving sessions, 41 percent are willing to charge once, 23 percent twice, and the remaining 12 percent are willing to charge three or more times per driving session (Sanguinetti and Kurani 2020). Today, many cities have relatively undeveloped or uncoordinated public charging infrastructure. In California, public charger access is lower in low-income and minority communities (Hsu and Fingerman 2021). Despite the small share of EVs among vehicles used for ride-hailing today, there is evidence that electric ride-hailing has already led to increased use of public chargers and has even saturated charging stations in some markets (Jenn 2019), meaning that significant infrastructure investments will be needed to scale up electric ride-hailing (Hall and Lutsey 2020; Nicholas et al. 2020; Schroeder et al. 2021).

Time has a cost. Time spent charging an EV could mean forgoing trips and earnings, making EVs less attractive (Nicholas et al. 2020; Slowik et al. 2019b). Among U.S. ride-hailing drivers who do charge during their driving sessions, the total time spent per charging stop (including traveling to a public charger, waiting to access the charger, charging the vehicle, and traveling to their next fare) ranged from 103 to 293 minutes. However, ride-hailing drivers with EVs do not seem to experience significant additional downtime or lost earnings. EVs used for ride-hailing in Los Angeles averaged only 23 minutes of additional downtime per day (6 percent) compared to ICE vehicles used for ride-hailing (Schroeder et al. 2021). Most U.S. ride-hailing drivers (54 percent) report that they do not miss out on earnings due to charges; 24 percent report that they miss out on earnings once every 5 charges; 10 percent, once per 10 charges; and 12 percent, once per 15 charges or less (Sanguinetti and Kurani 2020). In addition, some drivers may not be fully aware of charging options (Jenn 2019). Improving access to and information about public charging would lower this time cost, strengthening the case for EVs for ride-hailing.

Meanwhile, ride-hailing can improve the early business case for charging stations, which entail high up-front investments with relatively few existing customers. Ride-hailing EV drivers use more fast charging than average EV drivers (Jenn 2019), and they can be contacted collectively through the ride-hailing app (for example, to make them aware of special charging incentives).

A general expansion of all EV charging infrastructure would disproportionately benefit ride-hailing drivers and may be a precondition to electrifying ride-hailing. However, given the large, growing literature on this topic, this section focuses on actions that more specifically relate to ride-hailing.

Actions to Address Barrier 2

Increase access to near-home overnight charging for drivers without private parking

ACTIONS BY RIDE-HAILING COMPANIES

- Integrate information on charging locations in ride-hailing apps, such as Didi did in China (Slowik et al. 2019a), and explore other features to support EV drivers, such as limiting route length to the battery's remaining range.
- Share resources with drivers on technical requirements, financing options, and legal considerations related to installing home charging equipment in their city. This could include resources that tenants could share with their landlords or property managers about installing EV charging infrastructure.

- Share anonymized data on areas that have large populations of ride-hailing drivers to inform the placement of chargers.
- Collaborate with parking facilities to offer overnight charging to drivers who live nearby.

ACTIONS BY GOVERNMENTS

Municipal governments could take the following actions:

- Together with utilities, explore the potential for pole-mounted chargers in residential areas because this can lower the cost of installing public on-street chargers (Bureau of Street Lighting n.d.).
- Create on-demand programs for residents to request the installation of charging stations on residential streets, such as in Amsterdam (Chenadec n.d.).
- With utilities, create a database of multiunit buildings with EV charging.

State and/or municipal governments (depending on the area) could take the following actions:

- Enact "charge by right" or "right to plug" mandates, requiring landlords to approve tenants' requests to install EV charging at their own expense. Examples include Spain's Right to Plug in the Horizontal Property Bill (Government of Spain 2019), Vancouver's Building Bylaw includes a provision that requires new multiunit residential buildings to provide EV charging stations for 20 percent of all parking stalls (Lopez-Behar et al. 2019b), and Colorado's revised statutes state that tenants can install Level 1 or 2 chargers at their own expense and landlords are prohibited from charging a fee for its placement or use.[12] Similar mandates have passed in Norway, Canada (Ontario), and the United States (California and several other states) (McLane et al. 2021; NESCAUM 2019).
- Offer incentives to owners or managers of multiunit buildings who install EV chargers for tenants, as exemplified in the Smart Columbus electrification program (McLane et al. 2021), Los Angeles's $4,000 rebate for multiunit building owners who install chargers (which rises to $5,000 in disadvantaged communities; LADWP n.d.a), or British Columbia's program where residents of multiunit buildings can jointly apply through their utility for rebates on the creation of a professional EV-ready building plan, upgrades to electrical infrastructure, and the purchase and installation of a charger (BC Hydro 2021).
- Create EV-ready building codes and ordinances, requiring a certain percentage of parking in public and private buildings to be ready for EV chargers, which will lower installation costs for vehicle owners (Lopez-Behar et al. 2019b). For example, Boulder County in Colorado requires new construction to have prewiring for chargers (McLane et al. 2021). Policies should avoid increasing parking capacity, which can incentivize vehicle ownership.
- Develop goals and plans for residential EV charging, incorporated into all relevant planning documents, including housing, zoning, electricity, and infrastructure plans.
- Offer incentives to developers who include EV charging in new housing, especially multiunit buildings, eventually requiring it for all new residential parking. Incentives could also be offered for EV charging in office buildings, where ride-hailing drivers might be able to charge overnight.

ACTIONS BY OTHER ACTORS

- Utilities can educate customers about programs and benefits related to EVs, such as is required in Colorado.[13]
- Private parking lots or garages, and owners or managers of multiunit dwellings or office buildings, could install EV charging equipment or enable drivers who rent spaces to do so. This could enable drivers without at-home parking to charge overnight close to home while utilizing excess parking capacity and expanding parking facilities' revenue stream and customer base.

Increase the number of public chargers or offer ride-hailing drivers preferential access

ACTIONS BY RIDE-HAILING COMPANIES

- Negotiate favorable access for ride-hailing drivers to use public charging stations, such as ride-hailing-only chargers or times or the ability to reserve chargers in advance, leveraging drivers' collective behavior and purchasing power. Examples include Uber's partnership with bp pulse, which operates fast charging hubs for fleet vehicles and offers access and discounts to ride-hailing drivers (bp 2021).
- Share electrification rates, projections, and targets with relevant governments and planning agencies to improve coordination in infrastructure planning, such as with the SharedStreets program (Marshall 2018).

- Partner with charging point operators (and potentially other ride-hailing companies) to develop charging stations with preferential ride-hailing pricing or access in locations that are in high demand by drivers or develop and/or operate ride-hailing-only charging stations directly (Grid Alternatives and EVNoire 2021).

ACTIONS BY GOVERNMENTS

- Invest in building out the charging network, ensuring there are enough public fast chargers (350 kilowatts) to support ride-hailing drivers.
- Make charging stations for publicly owned vehicles available to ride-hailing drivers or commercial vehicles when there is excess capacity.
- Include the ride-hailing use case in electric transportation and infrastructure plans.
- Incentivize public stations to offer preferential access to ride-hailing, commercial, or other high-mileage vehicles.

ACTIONS BY OTHER ACTORS

- Charging station operators (which can be utilities) can offer ride-hailing drivers favorable access, such as ride-hailing-only chargers or times or the ability to reserve chargers (Grid Alternatives and EVNoire 2021), such as the commitment by Shenzhen, China, to install over 5,000 dedicated taxi charging posts by 2020 (Hove and Sandalow 2019).
- Charging station operators can share data—such as aggregated charger usage patterns and anonymized real-time charging usage—with governments, utilities, and/or other ride-hailing companies to optimize the use of charging stations, including by ride-hailing drivers (Nicholas et al. 2020).

Barrier 3. Charging Costs: Some Ride-Hailing Drivers Lack Affordable EV Charging Options

Since at-home overnight charging is often the cheapest option, relying on public fast charging can increase an EV's TCO by 25 percent (Nicholas et al. 2020). Lower-income households pay more than higher-income households per kilowatt-hour (kWh) of electricity to charge their EV (Bauer et al. 2021), probably because they more often live in rented or multifamily housing without access to private parking for overnight charging (Nicholas et al. 2020).[14] Although rates for at-home charging are generally lower than purchasing gasoline, rates are often not optimized to promote equity or incentivize charging behavior that supports grid load management. Electricity rates for all types of charging are typically set by city or regional utility companies or commissions, so this barrier is one of the most likely to vary among cities. Charging costs can be one of the most complicated cost components for potential EV buyers to forecast.

Up-front costs for at-home charging equipment and installation can also pose a financial barrier. EV charging equipment can cost $200–$1,000 (Forbes 2021) and another $400–$4,100 for installation upgrades to outlets or chargers (Nicholas et al. 2020). A U.S. federal tax credit covers 30 percent of the purchase and installation costs of home chargers, up to $1,000 (Moloughney 2021), which can be combined with state-level incentives (DOE 2021b). British Columbia implemented policies in 2013 and 2017 that allowed multiunit residential building residents to apply for a 75 percent rebate on the cost of acquisition and installation of Level 2 charging stations (Lopez-Behar et al. 2019a).

A general redesign of public EV charging rates, such as revising utility's demand charges on charging stations to increase incrementally, would disproportionately benefit ride-hailing drivers and may be a precondition to the electrification of ride-hailing. However, a significant, growing literature exists on this topic (as reviewed in Chitkara et al. [2016]), so this section focuses on actions that more specifically target ride-hailing.

Actions to Address Barrier 3

ACTIONS BY RIDE-HAILING COMPANIES

Negotiate favorable pricing and access for ride-hailing drivers by leveraging their collective behavior and purchasing power by

- partnering with utilities, charging station operators, and charging equipment manufacturers to negotiate favorable pricing on at-home charging equipment and installation or favorable electricity pricing for at-home and public charging; for example, Uber is partnering with Enel X to offer drivers discounts on charging equipment (McLane et al. 2021); and
- partnering with utilities to include ride-hailing drivers in pilots for storage-as-a-service or real-time pricing programs.

Share and streamline information to

- increase drivers' awareness of charging prices and incentives in their areas; and

- enhance the ride-hailing app to help reduce charging costs, for example, by integrating features that show city-specific time-of-use rates or compare costs across nearby public charging stations. These features could build awareness among ICE vehicle drivers by, for example, pointing out charging stations on maps during trips.

Offer direct financial support to drivers

- for at-home charging equipment; and
- to help them gain long-term access to near-home overnight charging.

ACTIONS BY GOVERNMENTS

- Expand EV purchase subsidies to support costs of at- or near-home EV charging equipment, for both homeowners/tenants and building owners or managers.
- Subsidize charging costs for eligible low-income households as part of programs to increase the affordability of basic utilities.
- Incentivize or require utilities to assess the feasibility of variable electricity rates and vehicle-to-grid opportunities.
- State or municipal governments or utilities can provide clarifying documents that explain the process, available subsidies, and costs of home charger installation because these elements vary locally.

ACTIONS BY UTILITIES

- Implement variable electricity prices, such as time-of-use pricing or real-time pricing, which can be applied to both fast and overnight charging. This incentivizes customers to charge at off-peak hours; ride-hailing drivers are more likely to be able to charge during off-peak daytime hours than workday commuters. The price differential should be large enough to lead to behavior change but not so large as to incentivize more fast charging than the grid can handle, and it should be accompanied by educational materials about rates and potential savings. One step beyond time-of-use pricing would be real-time pricing, such as day-ahead or hourly pricing. This would better reflect the true price of energy and times of energy surplus or deficit. Fluctuating gasoline prices may have prepared customers for dynamic pricing of electric "fuel." San Diego Gas & Electric's Power Your Drive program developed an app to show day-ahead and hourly pricing for its workplace chargers, with positive user feedback (Chitkara et al. 2016; Fitzgerald and Nelder 2019; Myers 2018).
- Offer at-home charging equipment through a monthly service fee, which would help amortize costs and increase consumer flexibility and could include a "rent-to-own" option (Myers 2018).
- Provide a storage-as-a-service (vehicle-to-grid) offering, where utilities compensate vehicle owners for using their EV for electricity storage when it is not in use. This could lower net costs of ownership, which may be especially appealing to price-sensitive ride-hailing drivers, though creating significant financial benefits may depend on technological and infrastructure improvements (Myers 2018).

ACTIONS BY CHARGING STATION OPERATORS (WHETHER PRIVATE COMPANIES, PUBLIC AGENCIES, OR UTILITIES)

- Offer ride-hailing drivers favorable rates to lower drivers' charging costs and secure a stable customer base for charging operators.
- Offer lower rates at lesser-used charging stations, which would be especially attractive to ride-hailing drivers since they may be more price sensitive or travel to a wider variety of locations.
- Offer loyalty programs or pricing systems where prices gradually decrease for high-mileage drivers, lowering average charging costs for ride-hailing drivers (though rates should be carefully designed to avoid incentivizing extra driving).
- Post information about pricing and charger availability online in real-time, and possibly enable reservations on chargers, to enable drivers to better plan stops. Partner with ride-hailing companies to embed this in their app.

Barrier 4. Awareness and Access: Some Ride-Hailing Drivers Lack Awareness or Access to Information regarding EVs

Some ride-hailing drivers lack awareness or access to information related to EV ownership and use, including cost and availability of charging, range, safety, available models, the relative TCO of EVs and ICE vehicles, and available subsidies and incentives (Grid Alternatives and EVNoire 2021; Jenn 2019). In some cases, relevant information, especially city-specific information, may not be easily accessible: it may be dispersed among different websites, overly technical, or only in English (Milken Institute 2017). U.S. drivers in a 2021 focus group said that uncertainty was a major barrier to them considering an EV, but that they would be open to changing their minds based on new information (Grid Alternatives and EVNoire 2021). Since EV technologies and policies are rapidly evolving, drivers may not be aware of recent changes, such as improved range, falling prices, charger availability, or incentives (Grid Alternatives and EVNoire 2021). Since ride-hailing drivers tend to have lower incomes and more diverse backgrounds than today's average EV owner, they may not see themselves as a potential EV owner (Sanguinetti and Kurani 2020).

Another reason for this may be the low visibility of EVs and chargers in everyday life, which may keep drivers from considering EVs or seeking out information on them, especially in places with limited access to EV models that are suitable for ride-hailing (Bui et al. 2020). The barrier of vehicle access is particularly relevant in smaller cities and in Canada, where 97 percent of EV sales were imported vehicles (compared to 53 percent in the United States; ICCT 2019). Though Canada is a major auto manufacturer and market, its EV production lags behind peers, which could pose a barrier to electrifying ride-hailing in the country if it leads to fewer models available or higher prices due to import taxes (ICCT 2019). In California, however, low-income households do not have to travel significantly farther than high-income households to access dealerships with EVs (Muehlegger and Rapson 2018).

Actions to Address Barrier 4

Drivers with well-functioning, satisfactory vehicles are unlikely to purchase different ones regardless of the case for EVs; the decision point will come whenever they replace their existing vehicles (Grid Alternatives and EVNoire 2021). This is often an unexpected event, arising when an existing vehicle breaks down, so they may not have time to conduct research or wait for approvals on rebate applications (Milken Institute 2017). Most members of disadvantaged communities that bought an EV were already very or exclusively interested in an EV when they began their vehicle search (CVRP 2016). A sustained, long-term, and culturally specific approach will be needed to raise overall awareness so that drivers will consider an EV when the time comes for their next vehicle. Also, lower-income households tend to buy vehicles through online platforms or from junkyards, where there are currently few EVs available (Milken Institute 2017).

Improve the visibility of ride-hailing EVs

ACTIONS BY RIDE-HAILING COMPANIES

- Offer an EV test-drive program in disadvantaged communities so drivers can familiarize themselves with the vehicle's performance and nearby charging options (Grid Alternatives and EVNoire 2021).
- Create marketing campaigns highlighting electric ride-hailing progress and goals.

Improve communication and education regarding EVs through tailored communications.

- Make all EV-related communications materials available in multiple languages (Grid Alternatives and EVNoire 2021) and (for audio or video) delivered by a person suited to the target audience.
- Develop communications materials featuring ride-hailing drivers with EVs sharing their experiences of buying and using the vehicles to build a positive perception of EVs in a credible peer-to-peer format (Grid Alternatives and EVNoire 2021).
- Explore innovative channels to share information about EVs with drivers, such as YouTube channels of drivers with EVs. Conduct focus groups with drivers to identify effective approaches, tailored for different demographics (Grid Alternatives and EVNoire 2021).
- Compensate drivers to take a course on EVs for ride-hailing, possibly developed and/or delivered by drivers with EVs, available in a variety of languages and times.
- Inform drivers of vehicle cost calculators (building on tools like Uber's EV cost calculator guide (Uber n.d.c), and potentially even prepopulate fields based on drivers' typical mileage or vehicle model (Sanguinetti et al. 2020).

ACTIONS BY GOVERNMENTS

- Municipal: Create a one-stop-shop webpage with all relevant subsidies and programs in that city—including programs from higher levels of government and private companies, financing support for credit-constrained households, and dealerships that sell EVs (Milken Institute 2017)—such as the Austin Energy EV Buyer's Guide (Austin Energy n.d.).

ACTIONS BY OTHER ACTORS

- Vehicle manufacturers or dealerships can offer test-drive programs for ride-hailing drivers or other high-mileage drivers (Grid Alternatives and EVNoire 2021). Utilities could also be involved, with the incentive of increasing electricity sales from high-mileage drivers.
- Car dealerships could offer salespeople incentives to sell EVs (Milken Institute 2017).

3. CITY SCORECARDS: SNAPSHOTS OF 10 CITIES' TRANSITIONS TO ELECTRIC RIDE-HAILING

Phasing the Transition to Electric Ride-Hailing

Cities will pass through several phases as they move towards full electrification of ride-hailing (see Figure 2). In the short term, stakeholders should focus on removing the barriers described in this study, among others, which currently inhibit the shift to electric ride-hailing. This will lead to Milestone 1, when barriers to electric ride-hailing have been removed, meaning EVs are at least as attractive as ICE vehicles for ride-hailing. However, many ICE vehicles will remain in use for years, so actions in Phase 2 should focus on *accelerating* EV adoption while avoiding inequitable impacts on drivers who still face difficulties in purchasing an EV.

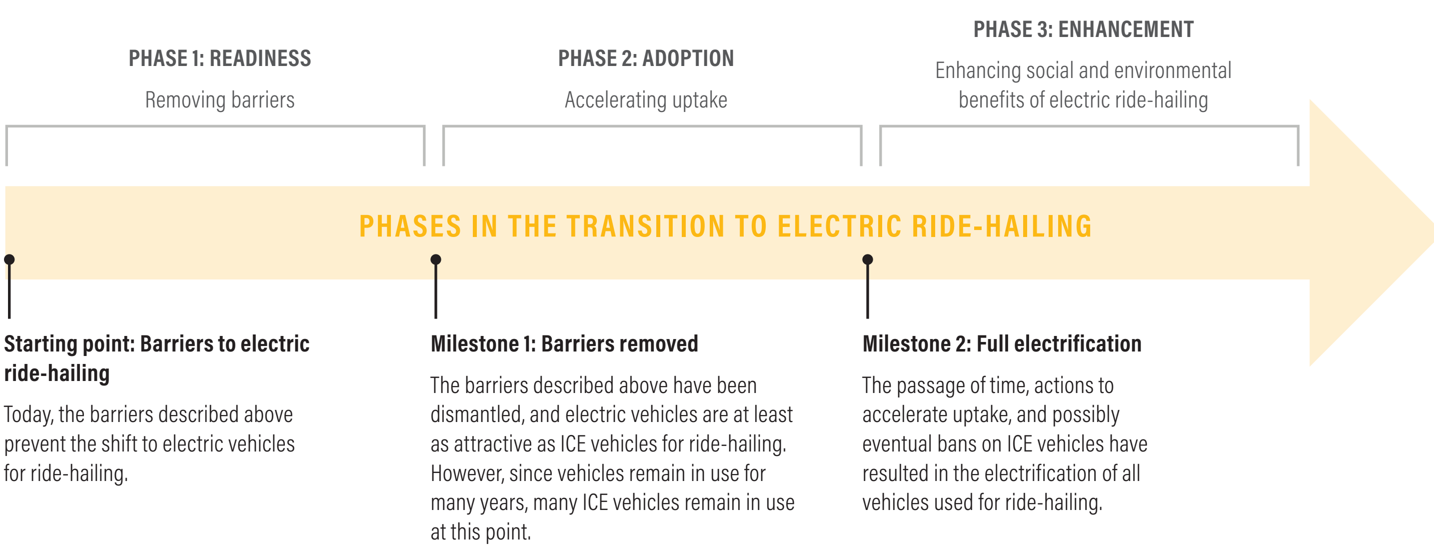
Throughout this process, there are opportunities to enhance the environmental, economic, and health benefits of electric ride-hailing. For example, smart charging can leverage EV batteries to smooth electricity demand and build resilience to power outages. Enhancing coordination with city planning and transportation agencies can help ride-hailing complement active and shared transit rather than adding vehicle miles. Policies can further decarbonize the electricity supply and promote a higher proportion of shared ride-hailing rides. Even after achieving Milestone 2—electrification of all vehicles used for ride-hailing—opportunities will remain to scale these efforts.

Today, essentially all cities are in Phase 1 of this framework. Cities are taking diverse approaches to dismantle barriers to vehicle electrification, and some are advancing faster than others. Although several analyses (including the one by Hall et al. [2020]) assess cities' overall progress towards electrifying transportation, little is known about progress specifically regarding electrifying vehicles used for ride-hailing.

These scorecards (Table 4) assess 10 U.S., European, and Canadian cities on the extent to which they have implemented measures to dismantle the four barriers to electric ride-hailing that are discussed in this paper (readiness) and their current levels of uptake of EVs for ride-hailing (adoption). These scorecards provide insight into city-level variation in readiness and adoption of electric ride-hailing and offer cities a template to explore their local barriers so they can tailor solutions. Although these scorecards focus on the city level, many important actions are in the remit of higher levels of government, utilities, or private companies.

See Appendix A for all data, assumptions, and calculations related to each indicator and each city. The color-coding criteria are detailed in Table 5.

Figure 2 | **Phases in the Transition to Electric Ride-Hailing**



Note: ICE = internal combustion engine.
Source: WRI authors.

Table 4 | City Scorecards: Ride-Hailing Readiness and Adoption

		Denver, Colorado, USA	Los Angeles, California, USA	New York City, New York, USA	Atlanta, Georgia, USA	Toronto, Canada	Vancouver, Canada	London, UK	Paris, France	Amsterdam, Netherlands	Berlin, Germany
READINESS	**Indicator 1: To what extent are EVs cost competitive with ICE vehicles that are commonly used by ride-hailing drivers?**										
	1a. Difference in cost of used ICE vehicles and used EVs commonly driven for ride-hailing, including applicable incentives	-$16,792	-$15,091	-$16,925	-$16,757	-$10,072	-$10,548	$3,888	-$13,390	-$5,930	$159
	1b. Difference in cost of a new EV purchased by a ride-hailing driver and a new EV purchased by an average EV buyer, including applicable incentives	-$7,500	-$3,000	-$7,500	-$7,500	$760	$0	$5,877	$2,272	$2,272	$6,816
	1c. Difference in cost of new ICE vehicles and new EVs commonly driven for ride-hailing, including applicable incentives	-$16,755	-$18,255	-$19,755	-$21,755	-$11,556	-$10,036	-$20,084	$750	-$2,370	$20,262
	1d. Incentives for purchasing used EVs	No	$1,500	No	No	$760	No	No	$1,136	$2,272	No
	Indicator 2: To what extent can ride-hailing drivers access EV charging?										
	2a. Public chargers per Uber EV	16 (2,400:150)	4 (9,600:700)	5 (2,100:400)	28 (2,700:100)	12 (2,100:200)	14 (1,300:100)	2 (5,500:3,000)	17 (4,700:300)	10 (2,800:300)	45 (1,300:30)
	2b. Preferential access to public chargers for EV ride-hailing drivers	Yes	No	No	No	No	No	Yes	Yes	No	No
	Indicator 3: To what extent can ride-hailing drivers charge EVs at an affordable price?										
	3a. Difference in cost per mile, gasoline vs fast charging	-$0.02	$0.02	-$0.02	-$0.02	$0.08	$0.09	$0.12	$0.12	$0.15	$0.10
	3b. Difference in cost per mile, gasoline vs at-home charging	$0.04	$0.05	$0.02	$0.04	$0.07	$0.09	$0.17	$0.17	$0.20	$0.15
	3c. Difference in cost per kWh, fast charging vs at-home charging	$0.24	$0.14	$0.15	$0.23	-$0.03	-$0.03	$0.17	$0.19	$0.18	$0.17
	3d. Financial support for at-home charger purchase and installation	$3,250	$2,000	$2,500	$1,250	$950	$1,863	$457	$1,761	$0	$0
	Indicator 4: To what extent can drivers access information on EVs and vehicle models suitable for ride-hailing?										
	4a. % EVs in city vehicle stock	0.1%	2.0%	0.1%	0.3%	0.4%	0.6%	1.2%	0.8%	7.4%	0.1%
	4b. Number of EV models available	21	43	33	22	26	26	96	118	119	130
	4c. City-level one-stop-shop EV information source	No	Yes	Yes	No	Yes	Yes	Yes	Yes	Yes	Yes
	4d. Do city EV policies, plans, or strategies account for ride-hailing?	Yes	Yes	Yes	No	Yes	No	Yes	Yes	Yes	Yes
	Indicator 5: How sustainable is the city's electricity supply?										
	5a. % of city electricity from renewable sources	21%	35%	8%	3%	37%	98%	12%	22%	18%	3%
ADOPTION	**Indicator 6: Adoption of EVs for ride-hailing**										
	6a. % EVs among vehicles used on Uber	1.1%	1.0%	0.6%	0.3%	0.4%	4.6%	5.3%	1.1%	6.7%	0.6%
	6b. % EVs among on-trip miles on Uber	0.8%	0.5%	0.5%	0.2%	0.3%	3.9%	5.6%	1.0%	6.3%	0.6%
	6c. Difference in % EV adoption, Uber vs city stock	1.0%	-1.1%	0.5%	0.0%	-2.1%	4.0%	4.1%	0.3%	-0.7%	1.4%

Notes: EV = electric vehicle; ICE = internal combustion engine; kWh = kilowatt-hour. Green shading indicates that the current conditions support the electrification of ride-hailing or that the indicator is likely on track for electrifying all vehicles used for ride-hailing by 2050. Yellow shading indicates that the current conditions offer insufficient support for the electrification of ride-hailing or that the indicator is moving in the right direction but is likely not on track. Red indicates that the current conditions are working against the electrification of ride-hailing or that the indicator is stagnant or moving in the wrong direction for electrifying all vehicles used for ride-hailing by 2050. All city data are from 2015 to 2021.

Sources: See Appendix A.

Table 5 | Criteria Used to Color-Code City Scorecards

Indicator 1: To what extent are EVs cost competitive with ICE vehicles that are commonly used by ride-hailing drivers?			
1a. Difference in cost of used ICE vehicles and used EVs commonly driven for ride-hailing, including applicable incentives	<-$3,000	-$3,000-$3,000	>$3,000
1b. Difference in cost of a new EV purchased by a ride-hailing driver and a new EV purchased by an average EV buyer, including applicable incentives	<$0		>$0
1c. Difference in cost of new ICE vehicles and new EVs commonly driven for ride-hailing, including applicable incentives	<-$3,000	-$3,000-$3,000	>$3,000
1d. Incentives for purchasing used EVs	<$0		>$0
Indicator 2: To what extent can ride-hailing drivers access EV charging?			
2a. Public chargers per Uber EV	0-5	6-15	>16
2b. Preferential access to public chargers for EV ride-hailing drivers	No		Yes
Indicator 3: To what extent can ride-hailing drivers charge EVs at an affordable price?			
3a. Difference in cost per mile, gasoline vs fast charging	<$0.02	$0.02-$0.05	>$0.05
3b. Difference in cost per mile, gasoline vs at-home charging	<$0.02	$0.02-$0.05	>$0.05
3c. Difference in cost per kWh, fast charging vs at-home charging	<$0	$0.00-$0.2	>$0.2
3d. Financial support for at-home charger purchase and installation	$0	$1-$1,500	>$1,500
Indicator 4: To what extent can drivers access information on EVs and vehicle models suitable for ride-hailing?			
4a. % EVs in city vehicle stock	<1%	1%-2%	>2%
4b. Number of EV models available	<30	30-40	>40
4c. City-level one-stop-shop EV information source	No		Yes
4d. Do city EV policies, plans, or strategies account for ride-hailing?	No EV policy	EV policy but no reference to ride-hailing	EV policy that includes ride-hailing
Indicator 5: How sustainable is the city's electricity supply?			
5a. % of city electricity from renewable sources	<10%	10-25%	>25%
Indicator 6: Adoption of EVs for ride-hailing			
6a. % EVs among vehicles used on Uber	<1%	1%-2%	>2%
6b. % EVs among on-trip miles on Uber	<1%	1%-2%	>2%
6c. Difference in % EV adoption, Uber vs city stock	<0%	0%-3%	>3%

Notes: EV = electric vehicle; ICE = internal combustion engine; kWh = kilowatt-hour. Detailed indicator descriptions are in Appendix A. Green shading indicates that the current conditions support the electrification of ride-hailing or that the indicator is likely on track for electrifying all vehicles used for ride-hailing by 2050. Yellow shading indicates that the current conditions offer insufficient support for the electrification of ride-hailing or that the indicator is moving in the right direction but is likely not on track. Red indicates that the current conditions are working against the electrification of ride-hailing or that the indicator is stagnant or moving in the wrong direction for electrifying all vehicles used for ride-hailing by 2050.

Source: WRI authors.

4. CITY SCORECARD TAKEAWAYS AND CONCLUSIONS

Although the scorecards are intended primarily to provide snapshots from 10 cities, this section highlights broader preliminary takeaways. In general, EVs are still more expensive than ICE vehicles in these cities. For used car purchases (Indicator 1a), in all included cities except London, a used EV costs more than or similar to (+/-$3,000) a comparable used ICE vehicle. For new EVs, in all U.S. cities, an average ride-hailing driver pays more than or the same as the current average EV buyer (1b), but in European and Canadian cities, an average ride-hailing driver pays a lower or similar price. This may reflect that, in the United States, ride-hailing drivers' lower incomes prevent them from taking full advantage of the federal tax credit, whereas European and Canadian subsidies might more effectively reach lower-income buyers. In U.S. cities, there is a smaller price gap between used ICE vehicles and EVs than between new ICE vehicles and EVs (1a and 1c), but in Amsterdam, Berlin, and Paris, new EVs are more cost competitive with new ICE vehicles; in Amsterdam and Berlin, however, this seems to mainly reflect national-level subsidies.

The number of chargers per Uber EV varied widely across cities, with no clear regional patterns (2a). London was unusual in having both a high number of chargers and the most EVs on Uber, leading to a low overall ratio. In contrast, Berlin had a high number of chargers per Uber EV, but mostly due to its low number of vehicles used on Uber. In terms of charging costs, at-home charging is cheaper than gasoline in all cities, but European and Canadian cities have charging rates that compare more favorably to fuel costs (3a and 3b).

All included cities are still far from electrifying both their ride-hailing and general vehicle stocks. U.S. cities are overall lagging in their rates of ride-hailing electrification (6a and 6b), with Amsterdam, London, and Vancouver leading. Berlin's low rate of electrification may reflect the impact of a German law that requires ride-hailing vehicles to return to a garage between rides (Frazzani et al. 2016). In most cities (besides Toronto), the share of EVs on Uber was similar to or greater than the share in the overall vehicle stock (6c); however, this may reflect a lack of recent data on the share of EVs in the city vehicle stock, compared to more up-to-date Uber electrification data. There was not significant variation between cities' share of EVs among Uber *vehicles* and share of EVs among Uber *on-trip miles* (6a and 6b), indicating that EVs used for ride-hailing are used with a similar intensity to ICE vehicles.

Ride-hailing can be electrified in a way that promotes equity and reduces emissions from transportation—if a range of actors overcomes the cost, charging, and access barriers described in this paper. Despite the promising first steps shown in the city scorecards, even bellwether cities have only begun the transition to electric ride-hailing. The potential actions in this paper can guide governments, ride-hailing companies, and others as they navigate the daunting, yet necessary, process of electrifying ride-hailing.

APPENDIX A: CITY SCORECARD DATA SOURCES AND METHODOLOGY

Table A1 | **City Scorecard Data Sources**

	INPUTS USED TO CALCULATE THE INDICATOR	DENVER, COLORADO, USA	LOS ANGELES, CALIFORNIA, USA	NEW YORK CITY, NEW YORK, USA	ATLANTA, GEORGIA, USA
Indicator 1: To what extent are EVs cost competitive with ICE vehicles that are commonly used by ride-hailing drivers?					
1a. Difference in cost of used ICE vehicles and used EVs commonly driven for ride-hailing, including applicable incentives	Used vehicle prices	KBB n.d.b	KBB n.d.b	KBB n.d.b	KBB n.d.b
1b. Difference in cost of a new EV purchased by a ride-hailing driver and a new EV purchased by an average EV buyer, including applicable incentives	EV purchase incentive (federal)	Fueleconomy.gov 2021	Fueleconomy.gov 2021	Fueleconomy.gov 2021	Fueleconomy.gov 2021
	EV purchase incentive (state)	DENC 2021	PG&E 2021	NYSERDA 2021	N/A
	EV purchase incentive (city)	N/A	N/A	N/A	N/A
	EV purchase incentives for low-income buyers	N/A	PG&E 2021	N/A	N/A
	Scrappage incentive	N/A	BAR 2021	N/A	N/A
	Taxi/ride-hailing EV incentive	N/A	N/A	N/A	N/A
1c. Difference in cost of new ICE vehicles and new EVs commonly driven for ride-hailing, including applicable incentives	New vehicle prices	Manufacturer's website for country (see Table A2 for vehicle make/model)	Manufacturer's website for country (see Table A2 for vehicle make/model)	Manufacturer's website for country (see Table A2 for vehicle make/model)	Manufacturer's website for country (see Table A2 for vehicle make/model)
1d. Incentives for purchasing used EVs	Used EV financial support	N/A	LADWP n.d.b	N/A	N/A
Indicator 2: To what extent can ride-hailing drivers access EV charging?					
2a. Public chargers per Uber EV		DOE 2021a	DOE 2021a	DOE 2021a	DOE 2021a
2b. Preferential access to public chargers for EV ride-hailing drivers		Atlas Public Policy 2021	Atlas Public Policy 2021	Atlas Public Policy 2021	Atlas Public Policy 2021
Indicator 3: To what extent can ride-hailing drivers charge EVs at an affordable price?					
3a. Difference in cost per mile, gasoline vs fast charging	Gasoline cost	EIA 2021b	EIA 2021b	EIA 2021b	EIA 2021b
3b. Difference in cost per mile, gasoline vs at-home charging	Residential electricity cost	EIA 2021a	EIA 2021a	EIA 2021a	EIA 2021a
3c. Difference in cost per kWh, fast charging vs at-home charging	Fast charging cost	McLane et al. 2021	McLane et al. 2021	McLane et al. 2021	McLane et al. 2021
3d. Financial support for at-home charger purchase and installation	Home charger incentive (federal)	IRS 2007	IRS 2007	IRS 2007	IRS 2007
	Home charger incentive (state)	N/A	LADWP n.d.b	Go Electric Drive n.d.	Georgia Power n.d.
	Apartment (shared) charger incentive	N/A	N/A	NYSERDA 2021	N/A

Table A1 | **City Scorecard Data Sources (Cont'd)**

	TORONTO, CANADA	VANCOUVER, CANADA	LONDON, UK	PARIS, FRANCE	AMSTERDAM, NETHERLANDS	BERLIN, GERMANY
Indicator 1: To what extent are EVs cost competitive with ICE vehicles that are commonly used by ride-hailing drivers?						
1a. Difference in cost of used ICE vehicles and used EVs commonly driven for ride-hailing, including applicable incentives	KBB n.d.a	KBB n.d.a	Autovisual n.d.	Autovisual n.d.	Autovisual n.d.	Autovisual n.d.
1b. Difference in cost of a new EV purchased by a ride-hailing driver and a new EV purchased by an average EV buyer, including applicable incentives	Transport Canada 2021a	Transport Canada 2021a	Gov.UK n.d.	Entreprises n.d.	Nederland Elektrisch 2021	eMO n.d.
	Transport Canada 2021a	Transport Canada 2021a	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	Entreprises n.d.	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	IBB 2021
	Plug'n Drive 2021a	Scrap-It 2021	N/A	Métropole du Grand Paris 2021	Nederland Elektrisch 2021	eMO n.d.
	N/A	N/A	Gov.UK n.d.	N/A	N/A	IBB 2021
1c. Difference in cost of new ICE vehicles and new EVs commonly driven for ride-hailing, including applicable incentives	Manufacturer's website for country (see Table A2 for vehicle make/ model)	Manufacturer's website for country (see Table A2 for vehicle make/ model)	Manufacturer's website for country (see Table A2 for vehicle make/ model)	Manufacturer's website for country (see Table A2 for vehicle make/ model)	Manufacturer's website for country (see Table A2 for vehicle make/ model)	Manufacturer's website for country (see Table A2 for vehicle make/ model)
1d. Incentives for purchasing used EVs	Plug'n Drive 2021b	N/A	Wallbox 2021	Wallbox 2021	Wallbox 2021	Wallbox 2021
Indicator 2: To what extent can ride-hailing drivers access EV charging?						
2a. Public chargers per Uber EV	DOE 2021a	DOE 2021a	Mayor of London 2021	Hall and Lutsey 2020	Schmutz et al. 2021	Nicholas and Wappelhorst 2020
2b. Preferential access to public chargers for EV ride-hailing drivers	Atlas Public Policy 2021	Atlas Public Policy 2021	Hall et al. 2021	Hall et al. 2021	N/A	N/A
Indicator 3: To what extent can ride-hailing drivers charge EVs at an affordable price?						
3a. Difference in cost per mile, gasoline vs fast charging	Statistics Canada 2021	Statistics Canada 2021	Trading Economics 2021	Trading Economics 2021	Trading Economics 2021	Trading Economics 2021
3b. Difference in cost per mile, gasoline vs at-home charging	energyhub.org 2021	energyhub.org 2021	Eurostat 2021	Eurostat 2021	Eurostat 2021	Eurostat 2021
3c. Difference in cost per kWh, fast charging vs at-home charging	Plug'n Drive n.d.a	Plug'n Drive n.d.a	Ionity 2020	Ionity 2020	Ionity 2020	Ionity 2020
3d. Financial support for at-home charger purchase and installation	N/A	N/A	Wallbox 2021	Wallbox 2021	Wallbox 2021	Wallbox 2021
	N/A	ChargeHub 2021	Wallbox 2021	Wallbox 2021	Wallbox 2021	Wallbox 2021
	Transport Canada 2021b	ChargeHub 2021	Wallbox 2021	Wallbox 2021	Wallbox 2021	Wallbox 2021

Table A1 | **City Scorecard Data Sources (Cont'd)**

	INPUTS USED TO CALCULATE THE INDICATOR	DENVER, COLORADO, USA	LOS ANGELES, CALIFORNIA, USA	NEW YORK CITY, NEW YORK, USA	ATLANTA, GEORGIA, USA
Indicator 4: To what extent can drivers access information on EVs and vehicle models suitable for ride-hailing?					
4a. % EVs in city vehicle stock	Battery EV stock share	DOE 2016	DOE 2016	DOE 2016	DOE 2016
4b. Number of EV models available		Atlas Public Policy 2021	Atlas Public Policy 2021	Atlas Public Policy 2021	Atlas Public Policy 2021
4c. City-level one-stop-shop EV information source		N/A	PG&E 2021	NYSERDA 2021	N/A
4d. Do city EV policies, plans, or strategies account for ride-hailing?		City and County of Denver 2020	LACI 2019	EV Shared Mobility 2020	N/A
Indicator 5: How sustainable is the city's electricity supply?					
5a. % of city electricity from renewable sources		City and County of Denver 2021	LADWP 2019	CNCA n.d.	CDP Worldwide 2015
Indicator 6: Adoption of EVs for ride-hailing					
6a. % EVs among vehicles used on Uber		Uber	Uber	Uber	Uber
6b. % EVs among on-trip miles on Uber		Uber	Uber	Uber	Uber
6c. Difference in % EV adoption, Uber vs city stock	% EVs among vehicles used on Uber	Uber	Uber	Uber	Uber
	% EVs in city vehicle stock	DOE 2016	DOE 2016	DOE 2016	DOE 2016

Table A1 | City Scorecard Data Sources (Cont'd)

	TORONTO, CANADA	VANCOUVER, CANADA	LONDON, UK	PARIS, FRANCE	AMSTERDAM, NETHERLANDS	BERLIN, GERMANY
Indicator 4: To what extent can drivers access information on EVs and vehicle models suitable for ride-hailing?						
4a. % EVs in city vehicle stock	Dunsky Energy Consulting 2019	Government of BC 2021	Hall et al. 2020	Hall et al. 2020	Hall et al. 2020	Hall et al. 2020
4b. Number of EV models available	Statistics Canada 2018	Statistics Canada 2018	EV-volumes.com 2021	EV-volumes.com 2021	EV-volumes.com 2021	EV-volumes.com 2021
4c. City-level one-stop-shop EV information source	Transport Canada 2021a	Transport Canada 2021a	TfL n.d.b	Roulez Branchez n.d.	RVO 2021	eMO n.d.
4d. Do city EV policies, plans, or strategies account for ride-hailing?	Government of Toronto 2021	N/A	Autonomy 2021	Hall et al. 2021	Hall et al. 2021	Hall et al. 2021
Indicator 5: How sustainable is the city's electricity supply?						
5a. % of city electricity from renewable sources	CNCA n.d.	CNCA n.d.	CNCA n.d.	IRENA 2021	CNCA n.d.	Smee 2019
Indicator 6: Adoption of EVs for ride-hailing						
6a. % EVs among vehicles used on Uber	Uber	Uber	Uber	Uber	Uber	Uber
6b. % EVs among on-trip miles on Uber	Uber	Uber	Uber	Uber	Uber	Uber
6c. Difference in % EV adoption, Uber vs city stock	Uber	Uber	Uber	Uber	Uber	Uber
	Dunsky Energy Consulting 2021	Government of BC 2021	Hall et al. 2020	Hall et al. 2020	Hall et al. 2020	Hall et al. 2020

Notes: EV = electric vehicle; ICE = internal combustion engine; kWh = kilowatt-hour. Data for Indicator 6 provided by Uber in September 2021.

Methodology

Indicator 1: To what extent are EVs cost competitive with ICE vehicles commonly used by ride-hailing drivers?

This indicator describes the extent to which EVs are cost competitive with comparable models of ICE vehicles that are commonly used by ride-hailing drivers in each city. The up-front cost of the vehicle is determined by the manufacturer's retail price (or, in the case of used vehicles, the sale price) combined with financial incentives from various levels of government. We evaluated the price of both new and used vehicles. This analysis does not include vehicle taxes and fees, such as registration or permitting fees, or ownership costs such as insurance or maintenance.

In Indicator 1, the ICE vehicle and EV segments are represented by the three most popular models used for ride-hailing, which are listed in Table A2. The "most popular" models were defined as those with the most completed rides on the Uber platform in that country in October 2019. Uber provided the data to the researchers in the summer of 2021. October 2019 was chosen to represent the average because it is a month without any major holidays (which can disrupt travel and work patterns) before the onset of COVID-19 and related lockdowns. For countries that have multiple cities included in the city scorecards (the United States and Canada), the same vehicle makes and models are used for all cities in that country.

Accordingly, in the city scorecards and in this methodology, the term *ICE vehicles* refers to the average of the three ICE vehicle models listed above for that country, and *EVs* refers to the average of the three EV models listed above for that country.

The only exception is in the Netherlands, where the Volkswagen Touran is the second most popular ICE vehicle used on Uber but has been discontinued, so a new vehicle price was unavailable. Therefore, the average price of new ICE vehicles in the Netherlands is represented by the average of the first and third most popular vehicles.

Table A2 | **Top Three Most Popular EV and ICE Vehicle Makes and Models on Uber, by Country, October 2019**

COUNTRY	MAKE	MODEL	FUEL TYPE	RANK
Canada	Tesla	Model 3	EV	1
Canada	Nissan	Leaf	EV	2
Canada	Chevrolet	Bolt	EV	3
Canada	Toyota	Corolla	ICE	1
Canada	Honda	Civic	ICE	2
Canada	Hyundai	Elantra	ICE	3
France	Nissan	Leaf	EV	1
France	Tesla	Model S	EV	2
France	Tesla	Model 3	EV	3
France	Peugeot	508	ICE	1
France	Volkswagen	Passat	ICE	2
France	Mercedes-Benz	C-Class	ICE	3
Germany	Nissan	Leaf	EV	1
Germany	Hyundai	Ioniq Electric	EV	2
Germany	Tesla	Model 3	EV	3
Germany	Mercedes-Benz	E-Class	ICE	1
Germany	Škoda	Rapid	ICE	2
Germany	Mercedes-Benz	E 220	ICE	3
Netherlands	Hyundai	Ioniq Electric	EV	1
Netherlands	Nissan	Leaf	EV	2

Table A2 | **Top Three Most Popular EV and ICE Vehicle Makes and Models on Uber, by Country, October 2019 (Cont'd)**

COUNTRY	MAKE	MODEL	FUEL TYPE	RANK
Netherlands	Tesla	Model S	EV	3
Netherlands	Mercedes-Benz	E-Class	ICE	1
Netherlands	Volkswagen	Touran	ICE	2
Netherlands	Škoda	Octavia	ICE	3
United Kingdom	Nissan	Leaf	EV	1
United Kingdom	Tesla	Model S	EV	2
United Kingdom	Hyundai	Ioniq	EV	3
United Kingdom	Škoda	Octavia	ICE	1
United Kingdom	Mercedes-Benz	E-Class	ICE	2
United Kingdom	Volkswagen	Passat	ICE	3
United States	Tesla	Model 3	EV	1
United States	Chevrolet	Bolt	EV	2
United States	Tesla	Model S	EV	3
United States	Toyota	Camry	ICE	1
United States	Toyota	Corolla	ICE	2
United States	Honda	Accord	ICE	3

Source: Data for 2019 provided by Uber in June–September 2021.

To enable comparison across cities, all prices were converted to U.S. dollars using a 2016–20 average:

- 1 euro = $1.1360
- 1 British pound = $1.3060
- 1 Canadian dollar = $0.7603

1A. DIFFERENCE IN COST OF **USED** ICE VEHICLES AND **USED** EVS COMMONLY DRIVEN FOR RIDE-HAILING, INCLUDING APPLICABLE INCENTIVES

Description: This is the difference between the average net up-front cost of used ICE vehicles and used EVs.

Justification: Used vehicles were included because the majority of vehicles purchased by low-income households and driven for ride-hailing are not new (either purchased used or already owned for several years). Buyers of used vehicles miss out on a majority of EV financial incentives, which often only apply to new vehicles. Indicator 1c includes a very similar calculation for new vehicles.

Method:

Incentives: An Internet search and literature review were conducted to identify all incentives that are applicable to used EVs in each city (including tax credits, purchase rebates, subsidies for low-income buyers, subsidies for ICE-to-EV replacements, and so forth, from any level of government or other entity). These incentives were summed to calculate the total amount of incentive available to a buyer purchasing a used EV in each city.

Vehicle prices: For U.S. and Canadian cities, the prices of used vehicles are from the Kelley Blue Book website (accessed April 2021), based on a central zip code for each city, and other sources as listed for European cities. On the Kelley Blue Book website, a vehicle search was conducted using the following specifications: 2018 model (because the average vehicle is approximately three years old when it is first registered on the Uber platform), 30,000 miles (based on an average mileage of 10,000 per year), black, standard equipment, "good" condition (because this is the condition of the majority of vehicles sold on Kelley Blue Book), and private party value (which refers to sales directly between individuals rather than through a dealership because this a more common purchase process for low-income households). The following versions of vehicles were used: Toyota Camry LE Sedan 4D, Toyota Corolla LE Sedan 4D,[15] Tesla Model 3 Long Range Sedan 4D (Tesla 2021), Chevrolet Bolt EV LT Hatchback 4D (Chevrolet 2021); each is the midrange version of the make and model. Where other options were given, defaults were used wherever possible. For European cities, the prices of used vehicles are from the vehicle valuation tool on Autovisual (n.d.), specifying 48,000 kilometers (roughly equal to 30,000 miles), and the same specifications as above wherever possible.

Calculating net price difference: The sum of per-vehicle incentives in each city was subtracted from the average (*gross*) price of a used EV and ICE vehicle in each city to determine the average *net* price of a used EV and ICE vehicle in each city. Then, the average net used EV price was subtracted from the average net used ICE vehicle price to determine the difference in the average net up-front cost between used ICE vehicles and used EVs. This number is included in the scorecard. A negative number indicates that the average net up-front cost of EVs is higher than that of ICE vehicles.

Sources: Used vehicle prices in the United States and Canada were sourced from Kelley Blue Book (accessed April 2021). Used vehicle prices in Europe were sourced from the vehicle valuation tool on Autovisual (n.d.). The financial incentives were sourced from the government webpages at the federal, state, and city levels as well as TCO calculators such as the one offered by PG&E. Sources for each city's incentives are listed in Table A1.

1B. DIFFERENCE IN COST OF A NEW EV PURCHASED BY A RIDE-HAILING DRIVER AND A NEW EV PURCHASED BY AN AVERAGE EV BUYER, INCLUDING APPLICABLE INCENTIVES

Description: The metric captures the difference between the net cost of a new EV purchased by someone with the demographic characteristics of an average ride-hailing driver (or available specifically to ride-hailing drivers) and the net cost of a new EV purchased by someone with the demographic characteristics of today's average EV buyer.

Justification: This metric shows the extent to which current EV incentives are available to ride-hailing drivers and low-income households and the presence of incentives that specifically target these groups. The main demographic characteristic of ride-hailing drivers reflected here is income level; researchers considered an analysis based on where ride-hailing drivers live, but data were not available. The difference in access to incentives between ride-hailing drivers and typical EV buyers is due to two main factors: the inability of low-income households to fully utilize tax credits because of their relatively lower tax liabilities (which makes EVs relatively more expensive for them) and financial incentives that target low-income households or ride-hailing drivers (which makes EVs relatively cheaper for them).

Method: The method of calculating gross new vehicle prices is identical to that in Indicator 1a. We conducted an online search to identify financial incentives for new EV purchases that are available specifically to ride-hailing drivers or low-income buyers (with whatever definition the incentive program uses). We then subtracted the sum of incentives available to low-income buyers and ride-hailing drivers from the city's gross new EV price and subtracted the sum of incentives available to the general public (assuming the ability to fully take advantage of tax credits) from the city's gross new EV price. We then subtracted the former from the latter, resulting in the difference between the cost of a new EV purchased by a low-income buyer or ride-hailing driver and a new EV purchased by an average EV buyer, including applicable incentives.

Sources: The financial incentives available in each city are sourced from the government webpages at the federal, state, and city levels as well as TCO calculators such as the one offered by PG&E (2021). Sources for each city's incentives are listed in Table A1.

1C. DIFFERENCE IN COST OF **NEW** ICE VEHICLES AND **NEW** EVS COMMONLY DRIVEN FOR RIDE-HAILING, INCLUDING APPLICABLE INCENTIVES

Description: This is the difference between the average net up-front cost of new ICE vehicles and new EVs.

Justification: The relatively higher up-front cost of new EVs is often cited as a major barrier to electrification, and comparisons of the up-front cost of new EVs versus new ICE vehicles are commonly used to demonstrate the gap in affordability between the vehicle types (and the rate at which this gap is closing). This metric provides a more nuanced city-level comparison by calculating the difference in the up-front cost of an EV and an ICE vehicle in each city, including any locally applicable incentives.

Method: The method is identical to Indicator 1a, except that it includes only new vehicles and the incentives that are relevant to them. For U.S. cities, it is assumed that buyers would be able to take full advantage of the federal tax credit for new EV purchases. Indicator 1b includes a calculation that does not make that assumption and estimates incentives available to low-income buyers.

Sources: The prices of new vehicles are sourced from the vehicle manufacturer webpages for each country, as listed in Table A1. The financial incentives available to EVs in each city are sourced from the government webpages at the federal, state, and city levels as well as TCO calculators such as the one offered by PG&E. Sources for each city's incentives are listed in Table A1.

1D. INCENTIVES FOR PURCHASING USED EVS

Description: This metric indicates the availability of incentives to support the purchase of a used EV in each city.

Justification: Increasing the availability of used EVs will be a major component of making EVs more affordable and accessible. In some cases, or in the short term, support to expand the used EV market may be warranted, including financial incentives that apply to used EVs. This metric measures such efforts by governments.

Method: Where financial incentives exist, the dollar amount is included. The lack of any such incentives is indicated by "No." Examples of included incentives and policies include the Los Angeles Department of Water and Power's Charge Up LA! program, which offer rebates of up to $1,500 on the purchase of a qualifying used EV (LADWP n.d.b) and Ontario's CAN$1,000 incentive towards the purchase of a used EV (Plug'n Drive n.d.b.).

Sources: Government EV policy webpages at the federal, state, and city level. Sources for each city are listed in Table A1.

Indicator 2: To what extent can ride-hailing drivers access EV charging?

This indicator relates to Barrier 2 (charging access: some ride-hailing drivers lack convenient EV charging options). Charging availability is measured by the number of public fast chargers relative to the number of EVs used on the Uber app in each city, and whether ride-hailing drivers have preferential access to public chargers.

2A. PUBLIC CHARGERS PER UBER EV

Description: This metric compares the number of public chargers in the city to the number of EVs used on the Uber app.

Justification: The accessibility of public chargers is relevant for ride-hailing drivers who lack access to at-home charging as well as those who drive enough miles during one driving session to acquire additional fast charging. Sufficient availability of public chargers can thereby improve the experience of current ride-hailing EV drivers and create conditions for other drivers to switch to EVs.

Method: Public chargers are defined as being publicly accessible, operational chargers of any speed. The geographic boundary of each city is defined as the Uber market area for each city (as of summer 2021). For most included cities, this corresponds roughly with the city's metropolitan area (not the administrative boundary). Maps of each city's market area are included below. For U.S. and Canadian cities, we generated a count of public fast charging plugs available in the Uber market area using the U.S. Department of Energy's Alternative Fuels Data Center (DOE AFDC) database, which lists all the public charging stations and plugs available at each station in North America categorized by the level of charging (Level 1, Level 2, DCFC) and location. We manually selected the market area boundary for each city on the DOE AFDC website, and it automatically generated a count of chargers in the selected area. For European cities, the number of charging plugs was based on the cited sources. If data was unavailable on the number of plugs, the number of charging stations was used.

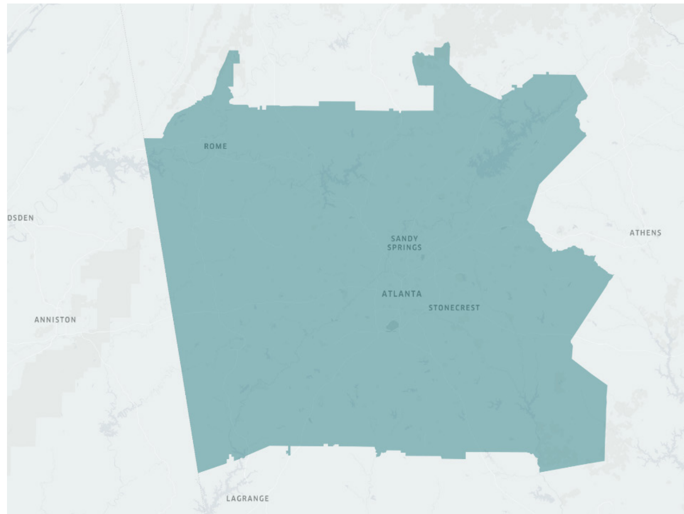
The number of ride-hailing EVs was calculated by multiplying the number of active Uber drivers (that have completed a minimum number of trips in a month within the Uber market area for the city) by the percentage of EVs in the vehicle count in the city's market area.

To calculate this metric, we divided the number of chargers in each city's market area by the number of EVs used on Uber in each city's market area. The resulting ratio was rounded to the nearest whole number. The number of EVs and the number of chargers are also listed, rounded to the nearest hundred and nearest multiple of ten, respectively.

Ideally, this metric would have compared the number of public chargers relative to the number of EVs used on all ride-hailing platforms, not just Uber. However, data were not available on the total number of ride-hailing EVs in each city. This approach assumes that EVs used for Uber represent a constant share of ride-hailing EVs across all cities, which is unlikely to be the case. Still, calculating this metric as a ratio enables some degree of comparison across cities because cities with a higher number of chargers per Uber EV may likely have a higher number of chargers per ride-hailing EV.

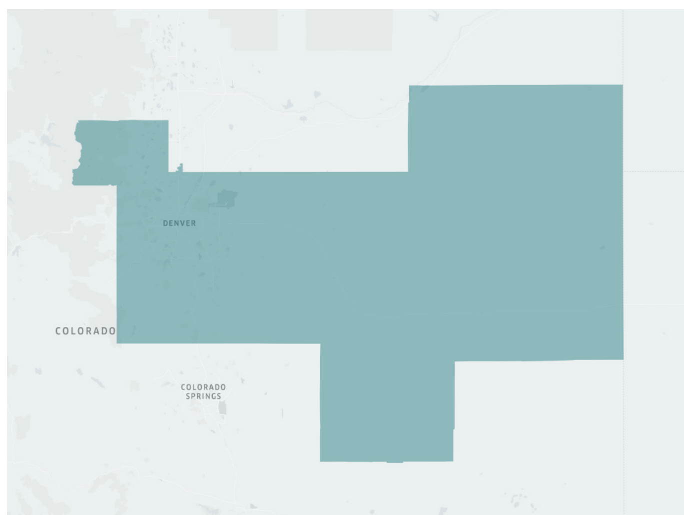
Uber market areas:

Figure A1 | **Atlanta, Georgia, USA**



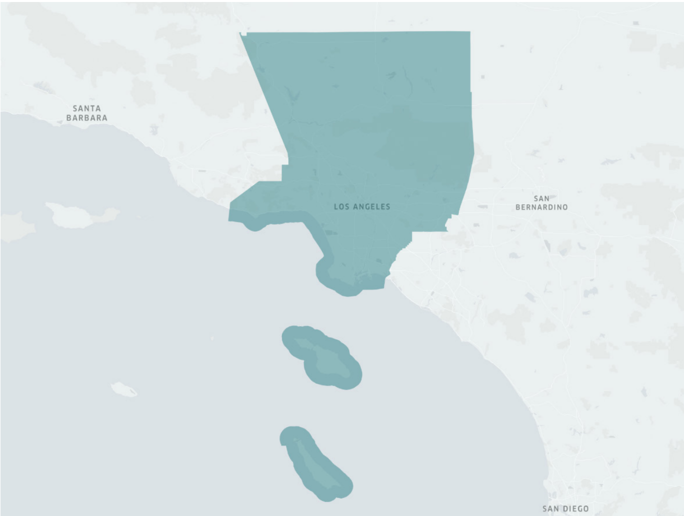
Source: Uber internal database, October 2021.

Figure A2 | **Denver, Colorado, USA**



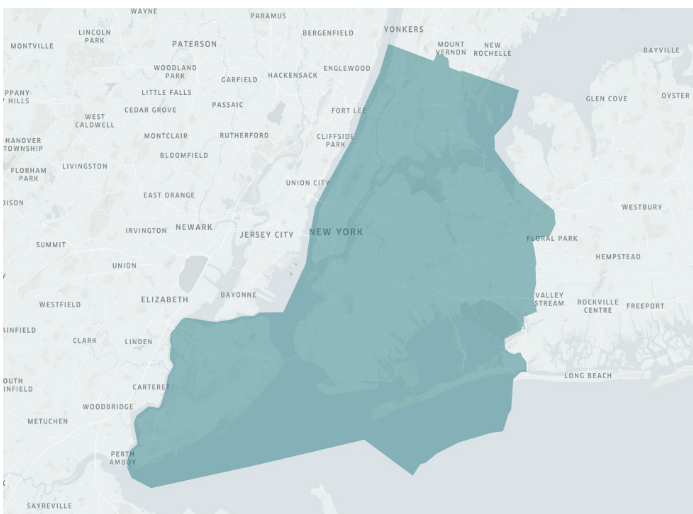
Source: Uber internal database, October 2021.

Figure A3 | **Los Angeles, California, USA**



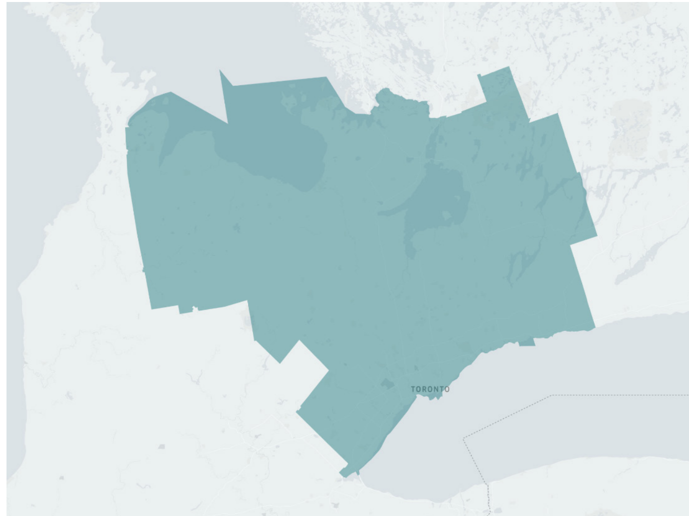
Source: Uber internal database, October 2021.

Figure A4 | **New York City, New York, USA**



Source: Uber internal database, October 2021.

Figure A5 | **Toronto, Canada**

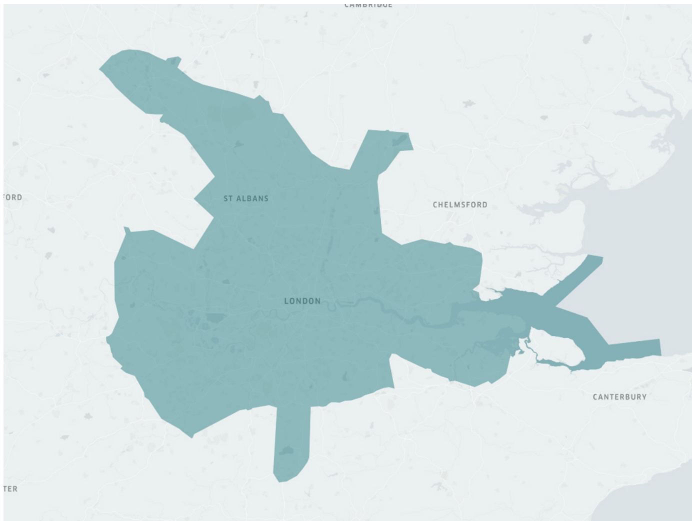


Source: Uber internal database, October 2021.

Figure A6 | **Vancouver, Canada**

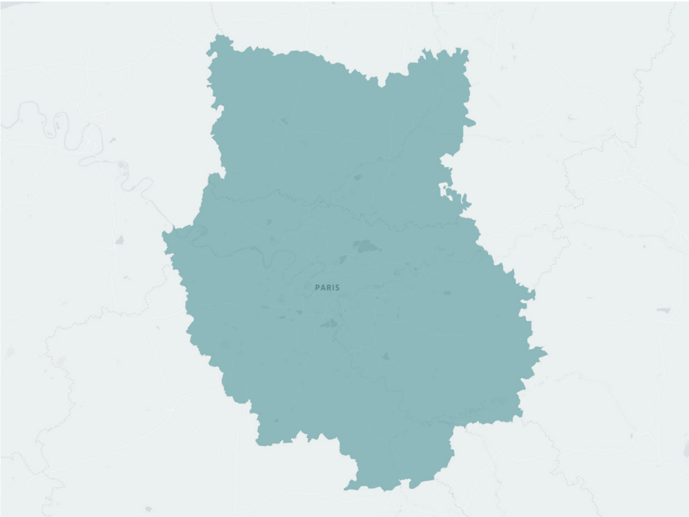
Source: Uber internal database, October 2021.

Figure A7 | London, United Kingdom



Source: Uber internal database, October 2021.

Figure A8 | Paris, France

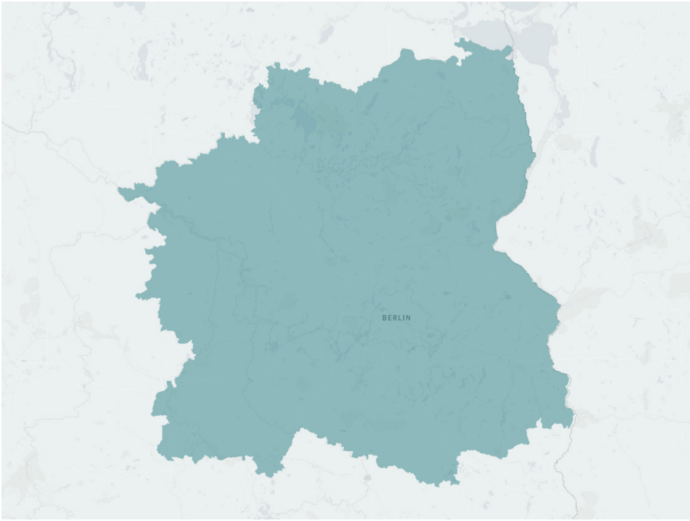


Source: Uber internal database, October 2021.

Figure A9 | Amsterdam, the Netherlands

Source: Uber internal database, October 2021.

Figure A10 | Berlin, Germany



Source: Uber internal database, October 2021.

Sources: Public charger availability data were sourced from the DOE AFDC database and Plugshare. Uber shared data on the number of ride-hailing drivers and share of EVs on the Uber app in each city with the researchers in the summer of 2021.

2B. PREFERENTIAL ACCESS TO PUBLIC CHARGERS FOR EV RIDE-HAILING DRIVERS

Description: This metric indicates the presence of measures that give preferential access to ride-hailing drivers at public charging stations.

Justification: This metric is important because ride-hailing drivers sometimes face a time cost from waiting to access public chargers. Giving ride-hailing drivers preferential access to strategically located public charging stations can reduce time spent charging.

Method: A "yes" in this category indicates the presence of business-to-business partnerships between ride-hailing companies and charging point operators or other relevant programs that give ride-hailing drivers (or a wider group that includes ride-hailing drivers) preferential access to public charging stations.

Sources: As listed in Table A1.

Indicator 3: To what extent can all ride-hailing drivers charge EVs at an affordable price?

This indicator relates to Barrier 3 (Charging costs: Some ride-hailing drivers lack affordable EV charging options). The affordability of EV charging depends on the residential electricity tariff, the cost of public fast charging, the fuel efficiencies of vehicles, the price of gasoline (including gas taxes), and the financial support available to offset the cost of purchasing and installing an at-home charger.

For the following metrics, fuel consumption for each vehicle is based on average fuel economies from the U.S. Environmental Protection Agency (Fueleconomy.gov 2021). Gasoline prices are averaged over 2015–20 to account for price fluctuations. Electricity tariffs are given for the most recent year available as electricity prices are relatively more stable.

3A. DIFFERENCE IN COST PER MILE, GASOLINE VS FAST CHARGING

Description: This metric indicates the difference in "fuel" (gasoline or electricity) costs per mile of travel from charging at public EV chargers compared to refueling at a gas station to travel one mile.

Justification: This metric uses the cost of public EV charging, which is especially relevant for ride-hailing drivers with high average mileage or those without access to overnight charging. Governments and businesses are experimenting with tariff structures, particularly the demand charges for fast charging events, so this metric reflects those initiatives. More affordable fast charging at public stations was assigned a better rating in the scorecards because it could increase the uptake of EVs for ride-hailing; however, in the long term, there will be better grid management (and possibly environmental) outcomes if the majority of charging is slow or "smart" charging rather than on-demand fast charging, meaning that fast charging should remain more expensive than overnight charging.

Method: We calculated the cost per mile of EV fast charging by dividing the city's DCFC tariff ($/kWh) by the average EV miles-per-gallon equivalent (MPGe), which is 131 according to Fueleconomy.gov as of August 2021. We calculated the gasoline cost per mile by dividing the city's five-year average gasoline price (or state-level data, when city data were unavailable) by 34, which is the combined city/highway miles per gallon of a 2018 Toyota Camry (2.5 liters, 4 cylinder, automatic [S8], regular gasoline), which is the most popular ICE vehicle used for ride-hailing in the United States. We then subtracted the EV cost per mile from the ICE cost per mile to calculate the difference between the costs of charging at a public fast charger and the cost of refueling at a gas station to travel one mile.

Sources: Fuel prices were sourced from the U.S. Energy Information Administration (EIA) and other sources as indicated for Canada and Europe. Electricity prices were sourced from the Rocky Mountain Institute, Plug'n Drive, and Ionity. Vehicle fuel efficiency data were sourced from the EIA.

3B. DIFFERENCE IN COST PER MILE, GASOLINE VS AT-HOME CHARGING

Description: This metric indicates the difference in "fuel" (gasoline or electricity) costs per mile of travel from charging an EV at home compared to refueling at a gas station to travel one mile.

Justification: This metric looks specifically at the relative cost of charging an EV at home. This is important because a significant share of EV charging by ride-hailing drivers takes place at home, and the relatively lower cost of at-home charging (compared to gasoline) is a major driver of the economic case for EVs. Different utilities are experimenting with electricity rate design particularly, such as time-of-use or pricing, which is reflected in this metric.

Method: The metric was calculated as the difference between the costs of charging an EV at home and the cost of refueling at a gas station to travel one mile.

Sources: Fuel prices were sourced from the EIA in the United States and other sources as indicated in Canada and Europe. Electricity prices were sourced from the EIA, Statistics Canada, and Trading Economics.

3C. DIFFERENCE IN COST PER KWH, FAST CHARGING VS AT-HOME CHARGING

Description: This metric compares electricity costs for different EV charging methods: public fast charging and at-home charging.

Justification: Since ride-hailing EV drivers use more fast charging than average EV drivers, the affordability of fast charging is especially important for the adoption of EVs for ride-hailing. Since demand charges often substantially increase the cost of fast charging, different approaches are being taken across cities to redesign electricity rates to reduce the price burden of fast charging. This metric captures the effect of some of these efforts.

Method: To calculate the difference in electricity costs for fast versus at-home charging, we used the city's fast and at-home charging prices per kWh, as described in Indicators 3a and 3b. We subtracted the city's at-home price per kWh from the city's fast charging price per kWh. A negative number indicates that at-home charging is more expensive than fast charging.

Sources: Fast charging rates are sourced from the city government webpages, charge point operators, and the Rocky Mountain Institute.

3D. FINANCIAL SUPPORT FOR AT-HOME CHARGER PURCHASE AND INSTALLATION

Description: This metric sums the financial incentives available per charging plug to offset the cost of installing an EV charger at a home (including single-family homes and multiunit buildings) in each city.

Justification: The cost of purchasing and installing home EV charging equipment may pose a financial barrier to ride-hailing drivers purchasing an EV, or it may lead to reliance on higher-cost fast charging or reliance on Level 1 slow-charging at home, which may be insufficient for high-mileage drivers. Financial support (from governments, ride-hailing companies, utilities, or other actors) can offer financial support to help overcome this barrier. This metric compares the financial support available in each city.

Method: This metric is the sum of all available incentives, from any actor, per charger plug. For incentives for chargers in multiunit buildings, which can have multiple plugs for use by multiple drivers, we assume the charger is used by an average of four drivers, based on national data of electric vehicle supply equipment ratios (number of plugs at charging station).

Sources: Government webpages at the federal, state, and city levels as well as TCO calculators such as the one offered by PG&E.

Indicator 4: To what extent can drivers access information on EVs and vehicle models suitable for ride-hailing?

This indicator relates to Barrier 4 (Awareness and access: Some ride-hailing drivers lack awareness or access to information regarding EVs). Ride-hailing drivers' access to and awareness of EVs in a city is defined in terms of the prevalence of EVs in the city's overall passenger vehicle stock, the number of EV models available for purchase in the city, the ease of access to information, and support for EV ride-hailing in relevant city or regional policy and planning documents.

4A. PERCENTAGE OF EVS IN CITY VEHICLE STOCK

Description: This metric is the percentage of EVs in the city's overall passenger vehicle stock.

Justification: Though a rough approximation, this metric is intended to reflect the awareness or prominence of EVs in the city, which indicates the extent to which ride-hailing drivers might be aware of the option of EVs or see themselves as a potential EV driver. Survey data on attitudes towards EVs would have been preferable but were not available consistently at the city level.

Method: This metric is the percentage of BEVs in the city's overall passenger vehicle stock.

Sources: The DOE AFDC database, the International Council on Clean Transportation, and other sources as indicated.

4B. NUMBER OF EV MODELS AVAILABLE

Description: This metric indicates the accessibility of EVs to ride-hailing drivers.

Justification: Access to a wide range of EV models increases the likelihood that ride-hailing drivers will have access to an EV model suitable for ride-hailing (in terms of mileage, trunk space, etc.), which would enable EV adoption among vehicles used for ride-hailing.

Method: In the United States, this metric uses the number of unique EV models available in the metro (core-based statistical) areas of each city. In Europe and Canada, city-level data were not available, so this metric uses the number of EV models available in the country. We assume that all EV models available in the country will be available in the capitals or major cities that are included in the city scorecards.

Sources: Atlas Public Policy (EV Hub), Canada Energy Regulator, and EV-Volumes.com.

4C. CITY-LEVEL ONE-STOP-SHOP EV INFORMATION SOURCE

Description: This metric indicates the presence of an easy-to-use, one-stop information platform that includes information on subsidies and support available from all levels of government and other actors.

Justification: This metric reflects the cities' efforts to increase awareness of EVs within their populations, which is important because some potential EV buyers, including ride-hailing drivers, sometimes struggle to locate all the information and programs that could support their decision to purchase an EV.

Method: "Yes" indicates that a webpage or communication campaign exists, led by the local government, aiming to provide useful information to potential EV buyers, such as information about incentives available for EV purchase or leasing, TCO calculators to help buyers understand the cost savings related to EVs, EV models available, and other information on, for example, sustainability issues related to EVs.

Sources: Various, as listed in Table A1.

4D. DO CITY EV POLICIES, PLANS, OR STRATEGIES ACCOUNT FOR RIDE-HAILING?

Description: This metric measures the extent to which cities' relevant policies, plans, or strategies include consideration related to electric ride-hailing.

Justification: Cities that included electric ride-hailing in their relevant policies, plans, or strategies have laid the foundation for taking action to dismantle the current barriers to electrification and are more likely to implement a coherent, strategic set of actions to do so.

Method: We searched for city-level policies, plans, or strategies related to EVs, EV infrastructure, transportation/mobility, electricity, land use, and climate. We analyzed whether any of those documents included a mention of the electrification of ride-hailing. An example of a "Yes" answer is the Denver Electric Vehicle (EV) Action Plan, which includes plans to "partner with car share, taxi, ride-hailing and emerging transportation and mobility providers (e.g., restaurant delivery services) to incentivize drivers to use EVs in ridesharing applications and incentivize customers to select EVs" (City and County of Denver 2020).

Sources: Various, as listed Table A1.

Indicator 5: How sustainable is the city's electricity supply?

5A. PERCENTAGE OF CITY ELECTRICITY FROM RENEWABLE SOURCES

Description: This metric reflects the percentage of each city's electricity that comes from renewable sources.

Justification: The total greenhouse gas emissions from driving an EV are directly related to the carbon intensity of the city's electricity supply.

Method: We assume that the electricity used to charge EVs contains the same mix of sources as the city's overall mix. This metric includes hydropower, biomass, wind, solar, and geothermal and excludes nuclear energy.

Sources: Various, as listed Table A1.

Indicator 6: Adoption of electric vehicles for ride-hailing

EVs are defined as being battery electric and fuel cell only, excluding all other types, such as plug-in hybrid, hybrid electric, range extenders, and so forth.

6A. PERCENTAGE OF EVS AMONG VEHICLES USED ON UBER

Description: This metric is the percentage of EVs among all vehicles driven on Uber in the city's market area.

Justification: This metric indicates the effectiveness of the actions in Indicators 1–4 in increasing the share of EVs used for ride-hailing, although if actions were implemented recently, there may be a time lag until the effects of those actions are seen.

Method: The number of ride-hailing EVs is defined as the number of active drivers on Uber using EVs. Uber defines the active drivers for each city as the drivers who have completed the minimum number of trips in a month and are operating within the Uber market area associated with that city. The metric only includes data from Uber and assumes that other ride-hailing companies in the city will have similar electrification rates because they are drawing from the same pool of potential drivers and experiencing the same policy environment.

Sources: Uber provided the average for January–June 2021 for each city. This longer-term average would smooth out any anomalous weeks or variations between weekdays and weekends.

6B. PERCENTAGE OF EVS AMONG ON-TRIP MILES ON UBER

Description: This metric is the percentage of on-trip miles driven by EVs among all vehicle miles driven on Uber in the city's market area.

Justification: This metric indicates the effectiveness of the actions in Indicators 1–4 in increasing the share of EVs used for ride-hailing, although if actions were implemented recently, there may be a time lag until the effects of those actions are seen. Compared to the electrification rate of vehicles, the electrification rate of miles more accurately captures the environmental and air quality impacts associated with the current level of electrification.

Method: This metric is calculated as the number of on-trip miles driven by BEVs divided by the total on-trip miles driven by all vehicles. The metric only includes data from Uber and assumes that other ride-hailing companies in the city will have similar electrification rates because they are drawing from the same pool of potential drivers and experiencing the same policy environment.

Sources: Data were provided by Uber for each city. It is the average for January–June 2021. This longer-term average would smooth out any anomalous weeks or variations between weekdays and weekends.

6C. DIFFERENCE IN PERCENTAGE OF EV ADOPTION, UBER VS CITY STOCK

Description: This metric compares adoption of EVs among vehicles used on Uber in the city to the adoption of EVs in the overall stock of passenger vehicles in the city.

Justification: Given the benefits of electrifying intensively used vehicles (such as vehicles used for ride-hailing) at a faster pace than less intensively used passenger vehicles, this metric reflects the extent to which cities are successful in prioritizing the electrification of intensively used vehicles. The metric only includes ride-hailing data from Uber and assumes that other ride-hailing companies in the city will have similar electrification rates because they are drawing from the same pool of potential drivers and experiencing the same policy environment. This means that the difference among cities would remain similar, even if the actual share of ride-hailing EVs would increase if EVs on other platforms were included.

Method: This metric calculates the difference between Indicator 6a (percentage of EVs among vehicles used on Uber) and Indicator 4a (percentage of EVs in city vehicle stock).

Sources: Data on vehicles used on Uber are the same as used for Indicator 6a, and data on city vehicle stocks are the same as used for Indicator 4a.

APPENDIX B: ESTIMATING RIDE-HAILING DRIVERS' INCOMES

Data availability on wages and household income is very limited, and estimates are further complicated by a range of factors, including hours worked and other income sources. Drivers' wages vary based on the demand from riders and the supply of drivers, and drivers can work any number of hours they choose, including varying the hours worked day-to-day or even month-to-month. In addition, many drivers have other full- or part-time jobs. In 2015, 52 percent of driver-partners worked full-time on another job, 14 percent of driver-partners had a part-time job in addition to partnering with Uber, and 33 percent of driver-partners had no other job (Hall and Krueger 2018) or had other household members who may earn income (see Table B1; Hall and Krueger 2018; Peticca-Harris et al. 2020; Uber n.d.b). This means that it is difficult to define the average income of a driver on the Uber app and the average income of a household that owns a vehicle that is driven for Uber.

Survey data suggest the average household income of drivers on the Uber app in the United States, Europe, and Canada is likely between \$20,000 and \$50,000 per year (or local currency equivalent). According to Mischel (2018),

Table B1 | **Share of Uber Drivers Who Average Fewer than 10 Hours per Week Online**

CITY	SHARE OF UBER DRIVERS WHO AVERAGE FEWER THAN 10 HOURS/ WEEK ONLINE (%)
Atlanta	42
Denver	35
Los Angeles	28
New York City	14
Toronto	30

Source: Data for 2019 provided by Uber in June–September 2021.

they earn an average of $11.77 per hour, pretax, after deducting Uber fees and drivers' vehicle expenses. This is significantly less than the $32.06 average hourly compensation of private sector workers, and the $14.99 average hourly compensation of service occupation workers, the lowest-paid major occupation (Mischel 2018). An annual salary for a full-time driver at the $11.77 hourly rate would be approximately $24,500, assuming 40 hours per week for 52 weeks per year (authors' analysis). Data from Gridwise, a company that helps drivers maximize their earnings on Uber and Lyft, published the following data on earnings of drivers who used their app in U.S. cities in 2020 (which may not be a typical year due to COVID-19 lockdowns; see Table B2). The average hourly rate is $16.61, and the median is $16.46 (authors' analysis). An annual salary for a full-time driver at the $16.46 hourly rate would be approximately $34,200 (authors' analysis). Notably, hourly earnings for drivers do not seem to correlate closely with the cost of living in a city, nor with the level of sprawl or quality of public transit. It is unclear what is underlying these earning discrepancies; it may be related to city- or state-specific fees or to the Gridwise app or COVID-19.

In the United Kingdom, according to a 2018 survey of 1,000 drivers on the Uber app in London, drivers were overwhelmingly male immigrants from the lower half of the income distribution (Berger et al. 2019). Their self-reported median gross weekly income (including income from sources other than Uber, before deducting vehicle expenses) was £460 ($645, using exchange rates for March 31, 2018), considerably less than the £596 ($835) median gross weekly pay of London workers at the time (Berger et al. 2019). Researchers estimated that the median drivers on the Uber app in London earned about £11 ($15.42) per hour spent logged into the app, after deducting Uber's service fee and vehicle-related expenses (Berger et al. 2019). An annual salary for a full-time driver at the $15.42 hourly rate would be approximately $32,000, assuming 40 hours per week for 52 weeks per year (authors' analysis). Based on this (admittedly limited) set of data, the following analysis will assume the average household income of a driver on the Uber app in the United States, Europe, and Canada is between $20,000 and $50,000 (or local currency equivalent). However, given the above data that 64 percent of drivers in 2015 had other sources of income, these annual extrapolations should only be considered rough estimates.

Table B2 | **Uber/Lyft Driver Earnings by Market (after Deducting Service Fee, before Vehicle Expenses)**

MARKET	MEDIAN EARNINGS PER HOUR
Fresno	$25.65
New York City	$23.48
Minneapolis	$21.73
San Jose	$20.06
Portland	$19.90
New Jersey	$19.64
Bay Area	$19.44
Philadelphia	$19.38
Sacramento	$19.33
Seattle	$19.20
Pittsburgh	$19.05
Denver	$18.90
Boston	$18.77
Detroit	$18.35
Milwaukee	$17.93
Phoenix	$17.90
Baltimore	$17.84
San Diego	$17.50
Washington, DC	$17.44
Chicago	$17.24
Los Angeles	$17.24
St. Louis	$17.21
Louisville	$16.89
Austin	$16.51
Cincinnati	$16.46
Cleveland	$16.44
Kansas City	$16.42
Dallas–Ft. Worth	$16.27

Table B2 | **Uber/Lyft Driver Earnings by Market (after Deducting Service Fee, before Vehicle Expenses) (Cont'd)**

MARKET	MEDIAN EARNINGS PER HOUR
New Orleans	$16.18
Las Vegas	$16.17
Salt Lake City	$16.04
Memphis	$15.87
Nashville	$15.72
Indianapolis	$15.59
Tampa Bay	$15.06
Tucson	$14.81
Columbus	$14.79
Raleigh-Durham	$14.77
San Antonio	$14.65
Charlotte	$14.60
Jacksonville	$14.59
Houston	$14.57
Miami	$14.03
Tulsa	$13.72
Oklahoma City	$13.33
Orlando	$13.33
Omaha	$12.29
Honolulu	$6.33
El Paso	$5.38

Source: Gridwise 2020.

ABBREVIATIONS

AFDC	Alternative Fuels Data Center
BEV	battery electric vehicle
CO_2e	carbon dioxide equivalent
DCFC	direct current fast charger
DOE	U.S. Department of Energy
EIA	U.S. Energy Information Administration
EV	electric vehicle
ICE	internal combustion engine
kWh	kilowatt-hour
MPGe	miles-per-gallon equivalent
pmt	passenger miles traveled
TCO	total cost of ownership
TNC	transportation network company

ENDNOTES

1. Data for 2019 provided by Uber in June–September 2021.
2. Data for 2019 provided by Uber in June–September 2021.
3. Data for 2019 provided by Uber in June–September 2021.
4. Data for 2019 provided by Uber in June–September 2021.
5. Data for 2019 provided by Uber in June–September 2021.
6. Data for 2019 provided by Uber in June–September 2021.
7. See Li-Cycle, https://li-cycle.com/, and Redwood Materials, https://www.redwoodmaterials.com/.
8. See the ReCell Center, https://recellcenter.org/.
9. See the Global Battery Alliance, https://www.weforum.org/global-battery-alliance/home.
10. For more information, see Flexdrive, https://www.flexdrive.com/; Drive Sally, https://www.drivesally.com/; Buggy, http://www.joinbuggy.com/; and Uber, https://www.uber.com/.
11. Oregon Revised Statutes 94 2021, https://oregon.public.law/statutes/ors_94.762.
12. Colorado Revised Statutes 2016: Title 38, https://leg.colorado.gov/sites/default/files/images/olls/crs2016-title-38.pdf.
13. Colorado Senate Bill 19-077 (2019), https://leg.colorado.gov/sites/default/files/2019a_077_signed.pdf.
14. However, it is important not to oversimplify this dynamic. In recent decades, in both the United States and Europe, more lower-income households are located in suburban or peripheral areas, where housing costs are cheaper than in the urban core (Cooke and Denton 2015; Covington 2015; Hochstenbach and Musterd 2018; Salvati et al. 2018). Homeownership status may be an equally strong explanatory factor, but more analysis is needed (Grid Alternatives and EVNoire 2021).
15. For the Toyota Camry and Corolla, see https://www.toyota.com/.

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World Resources Institute is a global research organization that turns big ideas into action at the nexus of environment, economic opportunity, and human well-being.

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We envision an equitable and prosperous planet driven by the wise management of natural resources. We aspire to create a world where the actions of government, business, and communities combine to eliminate poverty and sustain the natural environment for all people.

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